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Submitted by

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Under the guidance of

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Under the

DEPARTMENT OF PHYSIOTHERAPY

SURESH GYAN VIHAR UNIVERSITY

(Accredited by NAAC with 'A+' Grade)

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“CORRELATION BETWEEN FEAR OF MOVEMENT (KINESIOPHOBIA) AND STAIR-CLIMBING ABILITY FOLLOWING UNILATERAL TOTAL KNEE REPLACEMENT: A CROSS-SECTIONAL STUDY”

A DISSERTATION SUBMITTED TO

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Under the

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I hereby declare that this dissertation entitled "**Correlation between Fear of Movement (Kinesiophobia) and Stair-Climbing Ability following Unilateral Total Knee Replacement: A Cross-Sectional Study**" is a bonafide and genuine research work carried out by me under the guidance of Dr. Sheenam Popli, Associate Professor, Department of Physiotherapy, Suresh Gyan Vihar University, Jaipur. This work has not been submitted earlier, in part or in full, for the award of any degree or diploma of this or any other university.

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CONTENTS

> *Right-click the list below in Microsoft Word and choose "Update Field → Update entire table" to refresh page numbers.*

LIST OF GRAPHS

Graph No.	Title of Graph
Graph 1	Gender distribution of participants
Graph 2	Distribution of participants by affected side
Graph 3	Distribution of participants by age group
Graph 4	Distribution of participants by BMI category
Graph 5	Mean TSK-17 score and mean SCT time
Graph 6	Distribution of participants by level of kinesiophobia
Graph 7	Stair negotiation pattern
Graph 8	Handrail support during the Stair Climb Test
Graph 9	Assistive device used for ambulation
Graph 10	Histogram of TSK-17 scores with normal curve
Graph 11	Histogram of SCT time with normal curve
Graph 12	Normal Q–Q plot of TSK-17 scores
Graph 13	Normal Q–Q plot of SCT time
Graph 14	Scatter plot showing the correlation between TSK-17 score and SCT time
Graph 15	Mean SCT time in low and high kinesiophobia groups

LIST OF TABLES

Table No.	Title
Table 1	Interpretation of TSK-17 scores
Table 2	Interpretation of the correlation coefficient
Table 3	Mean and standard deviation of demographic variables (n = 50)
Table 4	Distribution of participants by gender and affected side
Table 5	Distribution of participants by age group
Table 6	Distribution of participants by BMI category
Table 7	Distribution of participants by post-operative duration
Table 8	Descriptive statistics of TSK-17 score and Stair Climb Test time
Table 9	Distribution of participants by level of kinesiophobia
Table 10	Stair negotiation pattern, handrail support and assistive device
Table 11	Tests of normality (Shapiro–Wilk)
Table 12	Pearson correlation between TSK-17 score and SCT time
Table 13	Simple linear regression of SCT time on TSK-17 score
Table 14	Comparison of TSK-17 score and SCT time between low and high kinesiophobia groups (independent t-test)
Table 15	Comparison of TSK-17 score and SCT time by gender, affected side and stair pattern (independent t-test)
Table 16	Comparison of TSK-17 score and SCT time by handrail support and assistive device (one-way ANOVA)
Table 17	Correlation of age, BMI and post-operative duration with TSK-17 score and SCT time
Table 18	Association of kinesiophobia level with stair pattern and handrail support (chi-square test)
Table 19	Hypothesis test summary
Table 20	Master chart of all participants (n = 50)

LIST OF ABBREVIATIONS

Abbreviation	Full form
ADL	Activities of Daily Living
AMI	Arthrogenic Muscle Inhibition
ANOVA	Analysis of Variance
BMI	Body Mass Index
CBT	Cognitive Behavioural Therapy
CI	Confidence Interval
EMG	Electromyography
FAM	Fear-Avoidance Model
ICC	Intraclass Correlation Coefficient
IEC	Institutional Ethics Committee
KOOS	Knee Injury and Osteoarthritis Outcome Score
LEFS	Lower Extremity Functional Scale
OA	Osteoarthritis
OARSI	Osteoarthritis Research Society International
PROM	Patient-Reported Outcome Measure
ROM	Range of Motion
SCT	Stair Climb Test
SD	Standard Deviation
STROBE	Strengthening the Reporting of Observational Studies in Epidemiology
TKA	Total Knee Arthroplasty
TKR	Total Knee Replacement
TSK	Tampa Scale for Kinesiophobia
TUG	Timed Up and Go test
WOMAC	Western Ontario and McMaster Universities Osteoarthritis Index

CHAPTER 1. INTRODUCTION

INTRODUCTION

1.1 Background

Knee osteoarthritis (OA) is one of the most common, disabling and progressive musculoskeletal disorders affecting the ageing population worldwide. It is characterised by progressive degeneration of articular cartilage, subchondral bone sclerosis, osteophyte formation and low-grade synovial inflammation, which together produce pain, stiffness, deformity and a marked decline in quality of life. When conservative measures such as analgesic and anti-inflammatory medication, intra-articular injections, physiotherapy, weight management and lifestyle modification no longer provide adequate relief of pain or improvement in function, Total Knee Arthroplasty (TKA), also termed Total Knee Replacement (TKR), is regarded as the gold-standard surgical treatment[(1)].

Large numbers of TKRs are performed every year with the aims of relieving arthritic pain, correcting frontal-plane deformity and restoring the mechanical stability of the joint, and the demand for the procedure is projected to rise steadily as populations age[(2)]. From a biomechanical and engineering standpoint, contemporary implants and surgical techniques achieve very high rates of success: most patients have satisfactory radiological alignment, a stable prosthetic joint and a substantial reduction in pain at rest after surgery[(3)].

Despite these structural achievements, there is a well-recognised gap between surgical success and patient-perceived outcome. Approximately one in five patients report dissatisfaction after primary TKR[(4)], and a considerable proportion continue to experience functional limitations during the first post-operative year even after an uncomplicated procedure. Many of these individuals demonstrate lower physical performance than age-matched peers without knee pathology. Level walking usually recovers to a satisfactory degree early in rehabilitation, whereas more complex locomotor tasks, which demand greater neuromotor coordination, eccentric muscle control and multi-joint movement, often remain impaired for a prolonged period[(5)].

1.2 Stair Climbing after Total Knee Replacement

Ascending and descending stairs are among the most challenging, and psychologically taxing, activities that a patient must regain after TKR. The ability to negotiate stairs independently is far more than a physical milestone: it is a marker of independence that allows a person to move in and out of the home, access public spaces and transport, and live comfortably in multi-level dwellings[(6)].

Biomechanical studies show that stair ambulation requires substantially greater lower-limb joint excursion, peak joint loading and muscular effort than level walking. During stair

ascent the quadriceps and gluteal muscles must generate large concentric forces to raise the body weight against gravity from knee flexion angles that commonly exceed 90–100 degrees[7]. Stair descent presents a different challenge, demanding finely graded eccentric contraction of the quadriceps to brake and control the downward acceleration of the body's centre of mass. Even minor deficits in neuromuscular coordination or proprioception therefore become evident during descent, which places additional stress on the prosthetic knee and provokes discomfort. Because of these high mechanical demands, stair negotiation has been described as the activity that causes the greatest difficulty for patients after unilateral TKR and is closely linked to fear and dissatisfaction[8].

Healthy individuals negotiate stairs with a reciprocal (step-over-step) pattern, placing alternate feet on successive steps. Following unilateral TKR, many patients adopt a non-reciprocal step-to (step-by-step) pattern in which the same limb always leads and the other follows. The non-operated limb typically leads during ascent and the operated limb during descent, so that the replaced knee is spared the most demanding concentric and eccentric loading. Although this strategy protects the operated knee, it imposes asymmetric loads on the lumbo-pelvic-hip complex and the contralateral limb. Patients also rely heavily on handrails, which unload the lower limbs, improve balance and reduce the risk of falls. While handrail use provides essential stability early after surgery, prolonged dependence on it is indicative of incomplete rehabilitation[9].

1.3 Biomedical Determinants of Stair-Climbing Ability

Traditionally, impaired stair negotiation after joint replacement has been studied from a biomedical perspective. The literature has concentrated on physical determinants such as active knee range of motion, prosthetic alignment, implant design and, above all, quadriceps weakness[10]. Arthrogenic muscle inhibition (AMI), an ongoing reflex inhibition of the muscles surrounding a joint after trauma or surgery, contributes to profound and long-lasting quadriceps atrophy and weakness after TKR.

Structural impairment, stiffness and quadriceps weakness clearly play important roles in recovery. However, multivariate analyses have repeatedly shown that physical deficits explain only part of the variance in stair-climbing ability after TKR. Patients with similar quadriceps strength, prosthesis design and knee flexion range display strikingly different stair performance: some climb efficiently with a reciprocal pattern, whereas others move hesitantly and rigidly and depend entirely on the handrail. These observations expose the limitations of a purely biomedical approach and point to the role of other mediating factors[11].

1.4 The Fear-Avoidance Model

Contemporary rehabilitation science increasingly views recovery from major reconstructive joint surgery as a psycho-physical process and recognises the value of incorporating cognitive-behavioural concepts. The fear-avoidance model (FAM) has emerged as a leading framework for explaining how cognitive and emotional factors influence the transition from acute pain to persistent pain and disability[12–14].

Originally developed to explain why some individuals with low back pain become chronically disabled, the model proposes that the way a person interprets pain determines the path of recovery. Acute pain, such as the intense inflammatory pain that follows TKR, can lead the patient along one of two pathways. When the patient appraises the pain as a normal, predictable and temporary consequence of tissue healing, fear remains low and the patient *confronts* the situation by participating actively in progressive physiotherapy, adhering to home exercise and gradually resuming more demanding activities. This adaptive pathway supports optimal physical and structural recovery[15].

When the appraisal is negative, a self-reinforcing maladaptive cycle begins. Its starting point is **pain catastrophising**, an exaggerated negative cognitive and emotional orientation towards actual or anticipated pain. Patients who catastrophise interpret their post-operative pain as highly threatening, uncontrollable and a sign of serious structural failure, such as wound breakdown or prosthetic loosening. This cognitive magnification amplifies central processing of nociceptive input and lowers pain thresholds[16].

Pain catastrophising is a direct antecedent of **kinesiophobia**, defined by Kori, Miller and Todd (1990)[20] as “an excessive, irrational and debilitating fear of physical movement and activity resulting from a feeling of vulnerability to painful injury or re-injury”. The patient holds the belief that exercise, exertion or movement will cause re-injury, irreversible damage or more pain[17]. Within the FAM, kinesiophobia is expressed through avoidance and hypervigilance: patients avoid movements expected to cause discomfort, limit their engagement with exercise and withdraw from valued activities. The consequences are disuse, depression and disability. Lack of loading leads to rapid muscle atrophy, joint stiffness, soft-tissue contracture and reduced cardiovascular fitness[18]. This physical deconditioning, combined with the distress of lost independence, paradoxically increases disability and pain, appears to confirm the catastrophic beliefs, and locks the patient in a vicious cycle of fear and decline[19]. The model as applied to TKR is summarised in Figure 1.1.

Figure 1.1: The fear-avoidance model applied to recovery after TKR

Adaptive (confrontation) pathway	Maladaptive (avoidance) pathway
Post-operative pain after TKR	Post-operative pain after TKR
↓ Pain appraised as normal and temporary	↓ Pain catastrophising (pain seen as a sign of damage)
↓ Low fear of movement	↓ Kinesiophobia (fear of movement / re-injury)
↓ Confrontation: active participation in exercise and graded return to stairs	↓ Avoidance and hypervigilance: step-to pattern, handrail dependence, offloading of operated knee
↓ Recovery of strength, ROM and motor control	↓ Disuse, deconditioning, depression and disability
↓ Independent, reciprocal stair negotiation	↓ Poor stair-climbing ability, which reinforces pain and fear

1.5 Kinesiophobia after Total Knee Replacement

Kinesiophobia is no longer regarded as confined to chronic low back pain or fibromyalgia. It is highly prevalent in acute orthopaedic populations undergoing major joint reconstruction such as TKR. Pooled estimates suggest a prevalence of approximately 35% after TKR, with some studies reporting rates as high as 56.67%[21,22].

This high prevalence reflects a distinctive feature of TKR rehabilitation: the operated joint must be actively mobilised and loaded almost immediately to prevent arthrofibrosis and adhesions. For a patient who has not been psychologically prepared before surgery, bending and bearing weight on a freshly operated, swollen and painful knee is sufficient to provoke an intense fear response[23].

The effects of kinesiophobia are particularly evident during complex and demanding tasks such as stair negotiation. When a fearful patient approaches a staircase, the limiting factor may not be weak muscles but fear-driven motor control. During stair descent, fear tends to cause excessive activation of antagonists such as the hamstrings and gastrocnemius, producing co-contraction around the knee to brace it against an anticipated collapse, fall or pain. This muscular stiffening prevents the normal kinematics of descent, as the patient cannot allow the knee to flex freely, and results in a slow, hesitant swing phase. The patient relies on the handrail not because strength is insufficient but because of fear, and persistent avoidance prevents the re-learning of eccentric motor control[24].

1.6 Unilateral versus Bilateral TKR

The expression of kinesiophobia appears to be influenced by whether the patient has undergone unilateral or bilateral TKR. Comparative cross-sectional studies have found higher rates of kinesiophobia after unilateral than after bilateral TKR (34.1% versus 22.5%) [25].

This difference has an important biomechanical and behavioural explanation. A patient with unilateral TKR has a non-operated limb that is relatively strong and pain-free, and therefore becomes the preferred limb for demanding tasks; mechanical demands can simply be shifted onto it[26]. Such offloading is a powerful avoidance strategy: the patient never has to learn to load the operated knee eccentrically during stair descent because the non-operated limb can always be used instead. This behaviour reinforces the fear-avoidance cycle and allows kinesiophobia to become entrenched. Patients with bilateral TKR have no pain-free limb on which to rely and are compelled to load both knees. For this reason, the present study focuses specifically on patients with unilateral TKR.

1.7 Rationale for the Present Study

Although the role of psychological factors in self-reported outcomes after TKR is increasingly acknowledged, the relationship between kinesiophobia and objectively measured stair performance remains poorly defined in routine clinical practice. Few studies have linked the Tampa Scale for Kinesiophobia (TSK) with performance-based measures of stair negotiation[27].

Understanding this relationship is clinically important because conventional physiotherapy programmes that address only biomechanical impairments may overlook the cognitive component of disability. If a patient's difficulty on stairs is driven more by persistent fear of movement than by actual muscle weakness, quadriceps strengthening and joint mobilisation alone are unlikely to be sufficient; instead, a holistic approach that integrates cognitive strategies with physical rehabilitation is required[28]. The present study therefore sets out to quantify the association between kinesiophobia and stair-climbing ability in individuals in the early phase of rehabilitation after unilateral TKR. By establishing whether psychological characteristics are associated with functional mobility independently of the joint, this work aims to encourage clinicians to look beyond the knee to the person who moves it[29].

1.8 Scope and Delimitations of the Study

- The study is delimited to individuals aged 50–75 years who have undergone primary unilateral TKR and are 4 weeks to 6 months post-operative.

- Kinesiophobia is assessed with the TSK-17 and stair-climbing ability with the timed SCT; other psychological constructs (anxiety, depression, catastrophising, self-efficacy) are not formally measured.
- Participants are recruited from [name of hospital(s) / rehabilitation centre(s)] in Jaipur, Rajasthan, which may limit generalisability.

CHAPTER 2. AIM AND OBJECTIVE

AIM AND OBJECTIVE

2.1 Aim

To study the relationship between fear of movement (kinesiophobia) and stair-climbing ability in individuals following unilateral Total Knee Replacement.

2.2 Primary Objective

1. To determine the correlation between fear of movement (kinesiophobia) and stair-climbing ability following unilateral TKR.

2.3 Secondary Objectives

1. To assess the level of kinesiophobia in individuals following unilateral TKR.
2. To evaluate stair-climbing ability in individuals following unilateral TKR.
3. To analyse whether higher levels of kinesiophobia are associated with poorer stair-climbing performance after unilateral TKR.

CHAPTER 3. RESEARCH PROBLEM OF THE STUDY

RESEARCH PROBLEM OF THE STUDY

3.1 Need of the Study

TKR is one of the most frequently performed procedures for advanced knee OA and other degenerative joint diseases. Although it is highly effective in relieving pain and restoring alignment, many patients continue to experience difficulty with functional tasks such as climbing stairs, rising from a chair and walking on uneven ground. Of these, stair climbing is among the most difficult and the most important for independent living.

Rehabilitation after unilateral TKR conventionally centres on physical impairments: pain relief, restoration of range of motion, lower-limb strengthening and gait training. Recovery, however, does not depend on physical factors alone. Psychological factors, and kinesiophobia in particular, may strongly influence a patient's engagement in rehabilitation and in functional activities. Patients with high kinesiophobia are reluctant to undertake weight-bearing tasks and restrict their movement for fear of pain or injury.

Despite growing recognition of psychological influences in musculoskeletal rehabilitation, little attention has been given to the association between kinesiophobia and stair climbing after unilateral TKR. Existing research has concentrated on overall function and pain rather than on specific, high-demand tasks that are vital for independence and social participation. Moreover, very little research has been conducted in the Indian population, in whom cultural factors, lifestyle demands (such as floor-level activities and multi-storey housing without lifts), access to rehabilitation and beliefs about pain and movement may differ and may therefore influence recovery differently.

The findings of this study can help physiotherapists to identify patients with high kinesiophobia and to design rehabilitation programmes that address both the psychological and physical barriers to recovery. Early identification and management of kinesiophobia may lead to better stair-climbing performance and greater functional independence.

3.2 Significance of the Study

1. **Clinical significance:** The study will inform physiotherapists and other rehabilitation professionals about the influence of fear of movement on stair-climbing ability after unilateral TKR. Understanding psychological barriers will enable clinicians to develop more comprehensive, biopsychosocial rehabilitation programmes.
2. **Functional significance:** Stair climbing is an essential activity of daily living and a key determinant of functional independence after TKR. The findings will clarify how kinesiophobia relates to functional mobility after surgery.

3. **Research significance:** Evidence on the relationship between kinesiophobia and stair-climbing ability after unilateral TKR is limited, particularly in India. This study will add to the existing literature and provide a basis for future longitudinal and interventional research.

3.3 Statement of the Problem

Despite successful surgery, many individuals who have undergone unilateral TKR continue to have difficulty negotiating stairs. Whether, and to what extent, fear of movement is associated with this difficulty has not been adequately established. The present study addresses the following research questions:

1. Is there a relationship between fear of movement (kinesiophobia) and stair-climbing ability among patients with unilateral TKR?
2. Do patients with high fear of movement have greater difficulty in stair climbing after unilateral TKR?
3. What is the level of fear of movement among patients with unilateral TKR?
4. How does stair-climbing ability differ according to the level of fear of movement among patients with unilateral TKR?

CHAPTER 4. IDENTIFICATION OF RESEARCH GAP

IDENTIFICATION OF RESEARCH GAP

1. **Limited studies specifically correlating kinesiophobia with stair-climbing ability after unilateral TKR.** Most post-TKR rehabilitation research focuses on pain management, range of motion, muscle strength or general functional mobility; little has examined fear of movement specifically in relation to stair negotiation in unilateral TKR.
2. **Emphasis on general functional outcomes rather than task-specific activities.** Much of the literature examines walking speed, balance or composite outcome scores such as WOMAC or KOOS, while stair climbing has been relatively neglected.
3. **Few studies combine objective stair-climbing measurement with a validated kinesiophobia scale.** Earlier studies have mainly relied on questionnaires; few have combined a valid fear measure such as the TSK with an objective performance test such as the Stair Climb Test (SCT).
4. **Limited evidence from the Indian population.** Most available studies come from Western populations. Differences in culture, pain perception, rehabilitation facilities and living conditions may influence outcomes in Indian patients.

CHAPTER 5. HYPOTHESIS

HYPOTHESIS

Null hypothesis (H_0): There is no significant correlation between fear of movement (kinesiophobia) and stair-climbing ability following unilateral TKR.

Alternative hypothesis (H_1): There is a significant correlation between fear of movement (kinesiophobia) and stair-climbing ability following unilateral TKR.

CHAPTER 6. REVIEW OF LITERATURE

REVIEW OF LITERATURE

The following review summarises relevant studies published in the last ten years (2016–2026) on kinesiophobia, functional outcome and stair-climbing ability after total knee arthroplasty/replacement, presented in chronological order.

1. **Filardo et al. (2016)**[30] conducted a prospective study to examine whether kinesiophobia influences recovery time and final outcome after total knee arthroplasty. A total of 101 patients (70 women and 31 men; mean age 66 years) were evaluated 5 days after surgery, at 1, 6 and 12 months, and at a mean final follow-up of 3.2 years, using the Tampa Scale for Kinesiophobia (TSK), pain, range of motion, function, WOMAC and SF-12. Fear of pain, and to a greater extent avoidance of movement, were strongly correlated with acute post-operative pain perception and with recovery up to one year. Although the influence of kinesiophobia diminished over longer follow-up, patients with higher kinesiophobia tended to have a poorer final outcome, and women appeared to be more affected. The authors concluded that kinesiophobia should be identified early because it slows recovery after TKA.

2. **Rasnick et al. (2016)**[31] used gait analysis and musculoskeletal modelling to estimate knee joint loads and the forces of surrounding muscles during stair ascent in patients with total knee replacement compared with healthy controls. Five patients with TKR and five age-matched controls ascended an instrumented staircase. Peak knee joint loads did not differ significantly between the groups; however, the TKR patients generated lower quadriceps forces, particularly in the rectus femoris and vastus lateralis, and compensated with greater activity of the knee flexors. The study showed that, although joint loading during stair ascent may appear normal after TKR, the underlying muscle recruitment pattern remains altered, which may contribute to persistent difficulty and inefficiency during stair climbing.

3. **Güney-Deniz et al. (2017)**[32] investigated whether kinesiophobia affects early functional outcomes following total knee arthroplasty. Forty-six patients were assessed on the day of discharge using the TSK, the 2-minute walk test (2-MWT), the Timed Up and Go (TUG) test, pain intensity and knee flexion range of motion, and were divided into high and low kinesiophobia groups. The low-kinesiophobia group walked further, reported less pain and had greater knee flexion, while TUG scores did not differ. TSK scores correlated with 2-MWT distance ($r = -0.40$), pain ($r = 0.80$) and knee flexion ($r = -0.47$). Regression analysis showed that kinesiophobia explained 41% of the variance in walking distance, 47% in knee flexion and 60% in pain. The

authors recommended that clinicians consider kinesiophobia when planning exercise programmes after TKA.

4. **Filardo et al. (2017)**[33] performed a prospective multivariate analysis of psychological and physical factors influencing the outcome of total knee arthroplasty in 200 patients (134 women and 66 men; mean age 65.7 years) evaluated 12 months after surgery. Kinesiophobia was measured with the TSK and depression with the Beck Depression Inventory, while pre-operative pain, function, sex, age and body mass index were included as covariates. Kinesiophobia emerged as a factor influencing outcome after TKA independently of the other psychological and physical variables. Its negative effect was particularly evident in women, and it showed a synergistic interaction with depression in producing poorer surgical outcomes. The authors suggested that patients with high kinesiophobia should be identified and managed to improve the results of TKA.

5. **Brown et al. (2020)**[34] conducted a systematic review to determine the effect of kinesiophobia on function, pain and range of motion after primary total knee replacement, and to identify interventions that reduce it. Electronic databases, grey literature and trial registries were searched, and 13 studies of adult primary TKR patients assessed with the Tampa Scale for Kinesiophobia were included; all showed an adequately low risk of methodological bias. The review concluded that kinesiophobia negatively affects functional outcome for up to one year after surgery and that active range of motion is reduced for up to six months. Post-operative functional and psychological interventions, including cognitive behavioural approaches, appeared to reduce kinesiophobia, supporting their inclusion in rehabilitation.

6. **Lee et al. (2020)**[35] carried out a prospective cohort study to identify pre-operative physical factors that predict stair-climbing ability one month after total knee arthroplasty. Eighty-four patients undergoing primary unilateral TKA were assessed before surgery and one month afterwards using a stair-climbing test, the six-minute walk test, the TUG test and isometric knee strength. Post-operative stair-ascent performance was associated with pre-operative TUG performance and extensor peak torque of the operated knee, whereas stair-descent performance was associated with pre-operative stair-descent performance and age. The authors concluded that pre-operative balance and quadriceps strength of the operated knee influence early stair-climbing ability, and that pre-operative physical function should be considered when planning rehabilitation.

7. **Deniz et al. (2022)**[36] examined how kinesiophobia affects short- and long-term spatiotemporal gait parameters in individuals undergoing unilateral total knee arthroplasty. Seventy patients were evaluated one week before surgery and at 3 and 12 months after surgery using the TSK, the Lequesne index and gait analysis on a Win-Track platform. Kinesiophobia increased at three months compared with the pre-operative period and decreased effectively by 12 months. Significant negative correlations were found between kinesiophobia and gait parameters in the early post-operative period. The authors concluded that the effect of kinesiophobia on gait should be evaluated at several time points around surgery and addressed during the rehabilitation process.

8. **Murakami et al. (2024)**[37] compared the difficulty of stair ascent and descent after total knee replacement using objective indicators. Seventy-eight patients were assessed three months after surgery, and stair negotiation was evaluated in terms of movement pattern (reciprocal or step-by-step), need for handrail support and the time taken to negotiate four steps. Stair descent was more difficult than ascent on all three indicators: a greater number of participants used a step-by-step pattern and handrail support during descent, and descent took longer. The authors attributed this to the high demand for eccentric quadriceps control during descent and recommended that rehabilitation after TKR place greater emphasis on stair-descent training.

9. **Aleksić et al. (2024)**[38] conducted an observational study to identify predictors of kinesiophobia and its influence on early functional outcomes after total knee arthroplasty. Seventy-five patients were assessed, and kinesiophobia was measured with the TSK on the fifth post-operative day; 27% of patients had kinesiophobia. The strongest predictors were the composite pain score on the fifth post-operative day (OR 6.2), the composite pain score on the first post-operative day (OR 2.1) and the intensity of chronic pain before surgery (OR 1.4). Patients with kinesiophobia were more dependent, walked more slowly and showed poorer recovery of knee extension by day five. The authors concluded that recognising and managing kinesiophobia could improve early recovery after TKA.

10. **Liu et al. (2024)**[39] performed a systematic review and meta-analysis to evaluate the effect of cognitive behavioural therapy (CBT) on pain, knee function and psychological status after primary total knee arthroplasty. Seven randomised controlled trials involving 608 patients were included, comparing CBT with usual care. CBT produced a significant reduction in kinesiophobia during the early post-operative period compared with usual care. However, no statistically significant differences were found between the CBT and usual-care groups for post-operative pain

or knee function. The authors concluded that CBT is effective in reducing fear of movement after TKA and recommended further high-quality trials to establish its effect on functional recovery.

11. **Xie et al. (2024)**[40] analysed the factors influencing post-operative kinesiophobia in patients undergoing total knee arthroplasty and constructed a nomogram prediction model. This retrospective observational study included 354 patients who underwent TKA between December 2019 and December 2022. Family monthly income, method of anaesthesia, pain (NRS), knee function (HSS score), activities of daily living and anxiety (SAS score) were identified as influencing factors. The nomogram showed acceptable discrimination, with a C-index of 0.721 and an area under the curve of 0.748, and good agreement between predicted and observed outcomes. The authors suggested that the model could help clinicians identify patients at high risk of kinesiophobia and intervene early.

12. **Özdemir and Karazeybek (2024)**[41] conducted a descriptive and correlational study to determine kinesiophobia, patient mobility and the factors affecting them in patients who had undergone total knee replacement. The study included 205 patients treated in orthopaedic and trauma wards between 2019 and 2022, assessed using a descriptive characteristics form, the Tampa Scale for Kinesiophobia and the Patient Mobility Scale. The mean TSK score was 40.47 ± 7.01, indicating high kinesiophobia. Kinesiophobia showed a moderate positive correlation with the mobility score ($r = 0.463$), a negative correlation with education level ($r = -0.281$) and a weak positive correlation with age ($r = 0.228$). The authors concluded that older age and lower education were associated with greater fear of movement.

13. **Asaki and Anan (2025)**[42] reported the case of an 80-year-old woman with bilateral knee osteoarthritis who underwent left total knee arthroplasty and continued to experience functional impairment and fear when descending stairs despite post-operative rehabilitation. At the initial out-patient evaluation, 78 days after surgery, she had limited knee range of motion, muscle weakness and high anxiety and fear of movement, and surface electromyography showed abnormal activity of the semitendinosus and medial gastrocnemius. After physiotherapy twice weekly, range of motion and self-efficacy improved by day 118, but fear of movement and anxiety persisted, together with a prolonged swing phase during stair descent. The authors emphasised that psychological factors must be addressed alongside physical rehabilitation.

14. **Du et al. (2025)**[43] carried out a systematic review and meta-analysis to estimate the prevalence of kinesiophobia after total knee arthroplasty and to identify its influencing

factors. Eleven studies involving 4,039 patients were pooled, giving an overall prevalence of 35% (95% CI 27–44%). Prevalence varied according to age, education, income and place of residence. Post-operative pain (OR 2.313), low social support (OR 1.681) and negative coping strategies (OR 1.344) were the principal factors associated with kinesiophobia, and low self-efficacy, older age and low education were also identified. The authors concluded that kinesiophobia is common after TKA and recommended routine screening and targeted interventions for high-risk patients.

15. **Wen et al. (2026)**[44] investigated the dynamic relationship between joint function and kinesiophobia following total knee arthroplasty using a longitudinal design and cross-lagged analysis. A convenience sample of 242 patients from two tertiary hospitals in China was assessed with the WOMAC and the 11-item Tampa Scale for Kinesiophobia (TSK-11) at 1, 3 and 6 months after surgery. Joint function improved progressively while kinesiophobia declined, with significant differences between all time points. Using cross-lagged panel analysis, the authors examined how kinesiophobia and joint function influence each other over time, highlighting that fear of movement and functional recovery are closely interlinked during the first six months after TKA.

TABLE OF REVIEW OF LITERATURE

Author and year	Study title	Study type	Sample size	Outcome measure	Key findings	Scope of future study
Filardo et al., 2016	Patient kinesiophobia affects both recovery time and final outcome after total knee arthroplasty	Prospective cohort	101	TSK, pain, ROM, WOMAC, SF-12	Kinesiophobia (especially activity avoidance) correlated with acute post-operative pain and slower recovery up to 1 year; poorer final outcome in women	Interventions targeting kinesiophobia early after TKA and their effect on long-term outcome
Rasnick et al., 2016	Knee joint loads and surrounding muscle forces during stair ascent in patients with total knee replacement	Comparative laboratory (biomechanical) study	10 (5 TKR, 5 controls)	3-D motion analysis, force platforms, musculoskeletal modelling	Similar peak joint loads, but reduced quadriceps forces and increased flexor forces in TKR during stair ascent	Larger samples; relate altered muscle recruitment to psychological factors such as fear of movement
Güney-Deniz et al., 2017	Does kinesiophobia affect the early functional outcomes following total knee arthroplasty?	Cross-sectional	46	TSK, 2-MWT, TUG, pain, knee flexion ROM	High kinesiophobia linked to shorter walking distance, more pain and less flexion; TSK explained 41% of 2-MWT, 47% of ROM and 60% of pain variance	Longitudinal studies and evaluation of other functional tasks such as stair climbing
Filardo et al., 2017	Kinesiophobia and depression affect total knee arthroplasty outcome in a multivariate analysis of psychological and physical factors on 200 patients	Prospective observational (multivariate analysis)	200	TSK, Beck Depression Inventory, clinical scores at 12 months	Kinesiophobia independently predicted poorer 12-month outcome, especially in women, and interacted with depression	Trials of combined psychological and physical interventions for high-risk patients
Brown et al., 2020	The effects of kinesiophobia on outcome following total knee replacement: a systematic review	Systematic review	13 studies	TSK; function, pain, ROM	Kinesiophobia impairs function up to 1 year and ROM up to 6 months after TKR; functional and psychological interventions reduce it	High-quality trials of interventions; task-specific functional outcomes

Author and year	Study title	Study type	Sample size	Outcome measure	Key findings	Scope of future study
Lee et al., 2020	Preoperative physical factors that predict stair-climbing ability at one month after total knee arthroplasty	Prospective cohort	84	Stair-climbing test, 6MWT, TUG, isometric knee strength	Pre-operative TUG and quadriceps torque predicted stair ascent; pre-operative stair descent and age predicted descent at 1 month	Include psychological factors (e.g., kinesiophobia) as predictors of stair ability
Deniz et al., 2022	How does kinesiophobia affect short- and long-term gait parameters in individuals with total knee arthroplasty surgery?	Prospective observational	70	TSK, Lequesne index, spatiotemporal gait analysis	Kinesiophobia peaked at 3 months and declined by 12 months; negatively correlated with gait parameters early after surgery	Effect of kinesiophobia on more demanding tasks; interventions in the early post-operative period
Murakami et al., 2024	Comparison of difficulty in stair ascent and descent after total knee replacement	Cross-sectional	78	Stair pattern, handrail use, time for four steps	Descent more difficult than ascent; more step-by-step pattern, handrail use and longer time during descent	Identify factors (physical and psychological) underlying descent difficulty
Aleksić et al., 2024	Understanding kinesiophobia: predictors and influence on early functional outcomes in patients with total knee arthroplasty	Prospective observational	75	TSK, pain scores, functional independence, gait speed, knee ROM	27% had kinesiophobia; post-operative pain was the strongest predictor; kinesiophobia linked to dependency, slower gait and poorer knee extension	Longer follow-up; pain-management and educational strategies to prevent kinesiophobia
Liu et al., 2024	Effect of cognitive behavioral therapy on pain, knee function, and psychological status in patients after primary total knee arthroplasty	Systematic review and meta-analysis	7 RCTs (608 patients)	TSK, pain, knee function, psychological scores	CBT significantly reduced early post-operative kinesiophobia; no significant effect on pain or knee function	Larger, high-quality RCTs with longer follow-up and functional outcomes

Author and year	Study title	Study type	Sample size	Outcome measure	Key findings	Scope of future study
Xie et al., 2024	Analysis of influencing factors and construction of nomogram prediction model of postoperative kinesiophobia in patients undergoing total knee arthroplasty	Retrospective observational	354	TSK, NRS, HSS, ADL, SAS	Income, anaesthesia, pain, knee function, ADL and anxiety predicted kinesiophobia; nomogram AUC 0.748	External validation of the model in prospective, multi-centre cohorts
Özdemir & Karazeybek, 2024	Kinesiophobia, patient mobility and influencing factors in patients undergoing total knee replacement	Descriptive correlational	205	TSK, Patient Mobility Scale	Mean TSK 40.47 ± 7.01; kinesiophobia correlated with mobility (r = 0.463), education (r = −0.281) and age (r = 0.228)	Objective performance-based measures of mobility; interventions for older, less-educated patients
Asaki & Anan, 2025	Persistent fear and extended swing phase during stair descent following total knee arthroplasty: a case report	Case report	1	ROM, strength, PROMs for fear/anxiety and self-efficacy, surface EMG, stair-descent kinematics	Fear of movement persisted despite improved ROM and self-efficacy, with abnormal muscle activity and prolonged swing phase in stair descent	Studies in larger samples relating fear of movement to stair performance
Du et al., 2025	Prevalence and influencing factors of kinesiophobia after total knee arthroplasty: a systematic review and meta-analysis	Systematic review and meta-analysis	11 studies (4,039 patients)	TSK	Pooled prevalence 35%; pain, low social support and negative coping were key risk factors	Studies from diverse populations (including India) and standardised TSK cut-offs
Wen et al., 2026	Investigating the dynamic relationship between joint function outcomes and kinesiophobia following total knee arthroplasty	Longitudinal cohort (cross-lagged analysis)	242	WOMAC, TSK-11	Joint function improved and kinesiophobia declined from 1 to 6 months; the two were dynamically related	Objective performance measures and longer follow-up

CHAPTER 7. FRAMEWORK OF RESEARCH PROJECT

FRAMEWORK OF RESEARCH PROJECT

The framework below summarises the sequence in which the study was planned and conducted.

Permission taken from the Institutional Ethics Committee (IEC) and the authorities of the study setting before the commencement of the study
↓ Patients who had undergone unilateral TKR (4 weeks – 6 months post-operative) were invited to participate in the study
↓ Screening of participants based upon the inclusion and exclusion criteria after a comprehensive examination
↓ Explanation of the study procedure and written informed consent obtained
↓ Recording of demographic and clinical details on the assessment proforma
↓ Assessment of fear of movement using the Tampa Scale for Kinesiophobia (TSK-17)
↓ Assessment of stair-climbing ability using the timed Stair Climb Test (SCT), with observation of stair pattern, handrail use and assistive device
↓ Data entered into the master chart and statistically analysed
↓ Results discussed and conclusions drawn

CHAPTER 8. MATERIALS AND METHODS

MATERIALS AND METHODS

8.1 STUDY DESIGN – Observational, cross-sectional, correlational study.

8.2 STUDY SETTING – Participants were recruited from [name of hospital / orthopaedic OPD / physiotherapy department], Jaipur, Rajasthan.

8.3 STUDY DURATION – [__ months (month, year to month, year)].

8.4 STUDY POPULATION – Individuals who had undergone primary unilateral Total Knee Replacement.

8.5 SAMPLING TECHNIQUE – Non-probability convenience sampling.

8.6 SAMPLE SIZE – N = 50 participants.

The sample size was calculated using G*Power software version 3.1.9.7[45] for a bivariate normal correlation model (two-tailed), with an expected correlation coefficient of $\rho = 0.40$, $\alpha = 0.05$ and power $(1 - \beta) = 0.80$. The minimum sample size obtained was 47 participants; allowing for possible dropouts, 50 participants were included.

8.7 Eligibility Criteria

Inclusion criteria for participants

- Individuals who had undergone primary unilateral Total Knee Replacement.
- Male and female patients aged 50–75 years.
- Medically stable and able to ambulate independently, with or without an assistive device.
- In the post-operative rehabilitation phase (4 weeks to 6 months after surgery).
- Able to understand and follow verbal instructions.
- Willing to participate and to provide written informed consent.

Exclusion criteria

- Bilateral Total Knee Replacement (simultaneous or staged).
- Revision TKR, or post-operative complications such as infection or implant failure.
- Neurological disorders affecting balance or mobility (e.g., stroke, Parkinson's disease).
- Severe cardiopulmonary conditions limiting physical activity.
- Other lower-limb musculoskeletal conditions affecting stair climbing (e.g., hip fracture, severe ankle pathology).

- Cognitive impairment or psychiatric disorders interfering with assessment.

ELIGIBILITY CRITERIA FOR SITE – A study site with a standard flight of stairs with handrails, adequate space for assessment, and permission for conducting the study.

8.8 Outcome Measures

PRIMARY OUTCOME –

1. **TAMPA SCALE FOR KINESIOPHOBIA (TSK-17)** – A 17-item self-report questionnaire measuring fear of movement/(re)injury[20,46]. Each item is scored from 1 (strongly disagree) to 4 (strongly agree); items 4, 8, 12 and 16 are reverse-scored, giving a total score of 17–68. A score above 37 indicates high kinesiophobia[47]. The scale has acceptable internal consistency (Cronbach's $\alpha = 0.68–0.91$), good test–retest reliability (ICC = 0.77–0.99) and established construct validity in musculoskeletal populations.
2. **STAIR CLIMB TEST (SCT)** – A performance-based test recommended by OARSI for people with knee osteoarthritis and after joint replacement[48]. The time in seconds taken to ascend and descend a flight of stairs is recorded; a longer time indicates poorer stair-climbing ability. The test has high test–retest reliability (ICC = 0.90–0.99), construct validity and responsiveness to rehabilitation.

SECONDARY OBSERVATIONS – Stair negotiation pattern (reciprocal / step-by-step), handrail support (none / unilateral / bilateral) and assistive device used (none / single cane / elbow crutch / walker).

Table 1: Interpretation of TSK-17 scores

TSK-17 total score	Interpretation
17–37	Low kinesiophobia
38–68	High kinesiophobia

HARM – The study involved no invasive procedure. Participants were guarded throughout the SCT to prevent falls.

8.9 Instruments Used

- Assessment proforma, consent forms, notepad and pen
- TSK-17 questionnaire
- Standard flight of [12] stairs (step height approximately [17–18] cm) with handrails
- Digital stopwatch
- Weighing scale and stadiometer / measuring tape

- Chair for rest

8.10 Variables

INDEPENDENT VARIABLE – Fear of movement (kinesiophobia), measured using the TSK-17.

DEPENDENT VARIABLE – Stair-climbing ability, measured using the SCT (time in seconds).

DEMOGRAPHIC VARIABLES – Age, gender, body mass index (BMI), post-operative duration, affected side, handrail support, stair negotiation pattern and assistive device.

8.11 Blinding

Not applicable, as this was an observational study without intervention. To reduce measurement bias, the TSK-17 was completed before the SCT so that test performance could not influence questionnaire responses.

8.12 Statistical Analysis

Data were entered in Microsoft Excel and analysed using [statistical software, e.g., IBM SPSS Statistics version __]. Descriptive statistics were expressed as mean ± standard deviation (SD) for continuous variables (age, BMI, post-operative duration, TSK-17 score and SCT time) and as frequency and percentage for categorical variables (gender, affected side, kinesiophobia level, stair pattern, handrail support and assistive device).

Normality of the continuous variables was assessed using the Shapiro–Wilk test. As TSK-17 and SCT scores were normally distributed, the correlation between them was analysed using Pearson's product–moment correlation coefficient (r), with Spearman's rank correlation (ρ) as a sensitivity analysis. Simple linear regression was used to estimate the change in SCT time per unit change in TSK-17 score. The independent-samples t-test was used to compare SCT time between participants with low (≤ 37) and high (> 37) kinesiophobia, and between subgroups of gender, affected side and stair pattern; one-way ANOVA was used for handrail support and assistive device; and the chi-square test was used for associations between categorical variables. Effect sizes were expressed as Cohen's d[49]. The strength of correlation was interpreted according to Mukaka (2012)[50] (Table 2). The level of significance was set at $p < 0.05$.

Table 2: Interpretation of the correlation coefficient

Size of correlation (|r|)	Interpretation
0.90 – 1.00	Very high

Size of correlation (|r|)	Interpretation
0.70 – 0.90	High
0.50 – 0.70	Moderate
0.30 – 0.50	Low
0.00 – 0.30	Negligible

8.13 Operational Definitions

Total Knee Replacement (TKR): A surgical procedure in which the articular surfaces of the femur, tibia and (optionally) patella are replaced with prosthetic components. *Unilateral TKR* refers to replacement of one knee only.

Kinesiophobia: An excessive, irrational and debilitating fear of physical movement and activity resulting from a feeling of vulnerability to painful injury or re-injury[20]. In this study it is quantified by the total score on the 17-item Tampa Scale for Kinesiophobia (TSK-17); a score above 37 is considered high kinesiophobia.

Stair-climbing ability: The capacity to ascend and descend a flight of stairs safely and efficiently. In this study it is quantified as the time in seconds taken to complete the Stair Climb Test (SCT); a longer time indicates poorer ability.

Reciprocal stair pattern: Ascending or descending with alternate feet placed on successive steps (step-over-step).

Step-to pattern: Ascending or descending by placing both feet on each step before proceeding to the next (non-reciprocal).

Post-operative rehabilitation phase: The period from 4 weeks to 6 months after surgery during which the participant is receiving or has recently received physiotherapy.

CHAPTER 9. STROBE STATEMENT

STROBE STATEMENT

The study is reported in accordance with the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines for cross-sectional studies[51]. The flow of participants through the study is shown below.

Fill in the number of patients assessed for eligibility and excluded (with reasons) from your screening log.

Patients with unilateral TKR assessed for eligibility (n = [___])
↓ Excluded (n = [___]): not meeting inclusion criteria (n = [___]); declined to participate (n = [___]); other reasons (n = [___])
↓ Enrolled and assessed (n = 50): TSK-17 questionnaire and Stair Climb Test
↓ Incomplete data / excluded from analysis (n = 0)
↓ Included in final analysis (n = 50)

STROBE checklist (cross-sectional studies) – summary

Item	Recommendation	Reported in
Title and abstract	Indicate the design; informative abstract	Title page; Chapter 19
Background / rationale	Scientific background and rationale	Chapter 1
Objectives	State specific objectives and hypotheses	Chapters 2 and 5
Study design	Present key elements of design	Chapter 8.1
Setting	Setting, locations, dates of recruitment and data collection	Chapter 8.2–8.3
Participants	Eligibility criteria, sources and methods of selection	Chapter 8.7
Variables	Define outcomes, exposures and potential confounders	Chapter 8.10
Data sources / measurement	Methods of assessment for each variable	Chapters 8.8 and 10
Bias	Efforts to address potential sources of bias	Chapter 8.11
Study size	Explain how the study size was arrived at	Chapter 8.6

Item	Recommendation	Reported in
Statistical methods	Describe all statistical methods	Chapter 8.12
Participants (results)	Numbers at each stage; flow diagram	Chapter 9
Descriptive data	Characteristics of participants	Chapter 11
Main results	Estimates with precision (e.g., 95% CI)	Chapter 11
Limitations	Sources of bias or imprecision	Chapter 15
Interpretation / generalisability	Cautious overall interpretation	Chapters 12–13

CHAPTER 10. PROCEDURE

PROCEDURE

PROCEDURE TO BE CONDUCTED ON ALL PARTICIPANTS

- Participants were screened on the basis of the selection criteria by reviewing their medical records and through a brief interview and physical examination.
- The purpose and procedure of the study were explained in the participant's preferred language (English or Hindi), and written informed consent was obtained.
- Demographic and clinical data – age, gender, height, weight, BMI, post-operative duration, affected side and assistive device – were recorded on the assessment proforma (Annexure I).
- Each participant completed the TSK-17 questionnaire in a quiet room before any physical testing. For participants unable to read, the items were read aloud by the investigator without prompting, and responses were recorded.
- After at least 5 minutes of seated rest, the Stair Climb Test was explained and demonstrated, and one familiarisation trial was allowed.
- On the command "Go", the participant ascended a flight of [12] stairs, turned at the top and descended to the bottom as quickly as safely possible. Timing started on "Go" and stopped when both feet returned to the floor at the bottom. Use of the handrail and the usual assistive device was permitted.
- The investigator guarded the participant throughout. The total time (seconds), stair negotiation pattern (reciprocal or step-by-step), handrail support (none, unilateral or bilateral) and assistive device were recorded (Annexure III).
- The test was stopped immediately in the event of severe pain, dizziness, breathlessness, chest discomfort or unsafe performance. [State whether any adverse events occurred.]
- All data were coded and entered into the master chart (Chapter 18) for statistical analysis.

> *Insert photographs of the assessment here if available (e.g., Fig. 1 – Participant completing the TSK-17; Fig. 2 – Stair Climb Test: ascent; Fig. 3 – Stair Climb Test: descent with handrail support), with the participant's consent.*

CHAPTER 11. RESULTS

RESULTS

SUB-CHAPTER 1: DESCRIPTIVE ANALYSIS

1. Demographic data

A total of 50 participants (20 male and 30 female) who had undergone unilateral Total Knee Replacement were included in the study. The mean age of the participants was 63.50 ± 7.05 years (range 51–74 years), the mean BMI was 26.81 ± 3.87 kg/m², and participants were assessed at a mean of 13.44 ± 6.44 weeks after surgery (range 4–24 weeks). Table 3 gives the mean and standard deviation of these variables.

Table 3: Mean and standard deviation of demographic variables (n = 50)

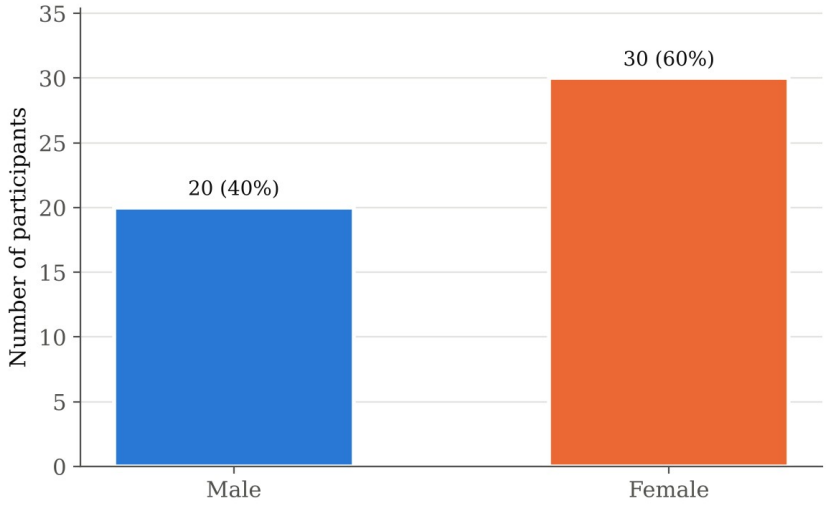
Variable	Mean	Std. Deviation	Minimum	Maximum
Age (years)	63.50	7.05	51	74
BMI (kg/m²)	26.81	3.87	20.1	37.2
Post-operative duration (weeks)	13.44	6.44	4	24

Table 4: Distribution of participants by gender and affected side

Variable	Category	Frequency (n)	Percentage (%)
Gender	Male	20	40.0
	Female	30	60.0
Affected side	Right	23	46.0
	Left	27	54.0
Total		50	100.0

Of the 50 participants, 30 (60.0%) were female and 20 (40.0%) were male. The left knee was operated in 27 (54.0%) participants and the right knee in 23 (46.0%) (Graphs 1 and 2).

Graph 1: Gender distribution of participants



Graph 2: Distribution of participants by affected side

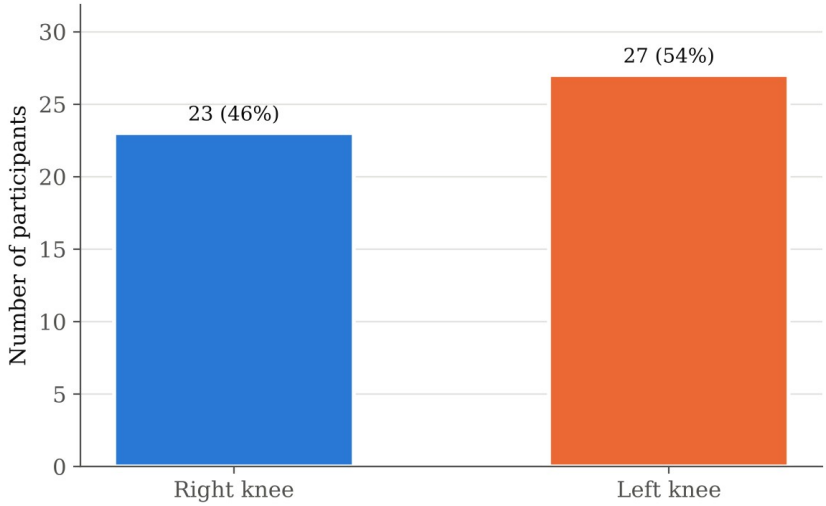


Table 5: Distribution of participants by age group

Age group (years)	Frequency (n)	Percentage (%)
50–59	17	34.0
60–69	19	38.0
70–75	14	28.0
Total	50	100.0

The largest proportion of participants was in the 60–69 year age group (19 (38.0%)), followed by 50–59 years (17 (34.0%)) and 70–75 years (14 (28.0%)) (Graph 3).

Graph 3: Distribution of participants by age group

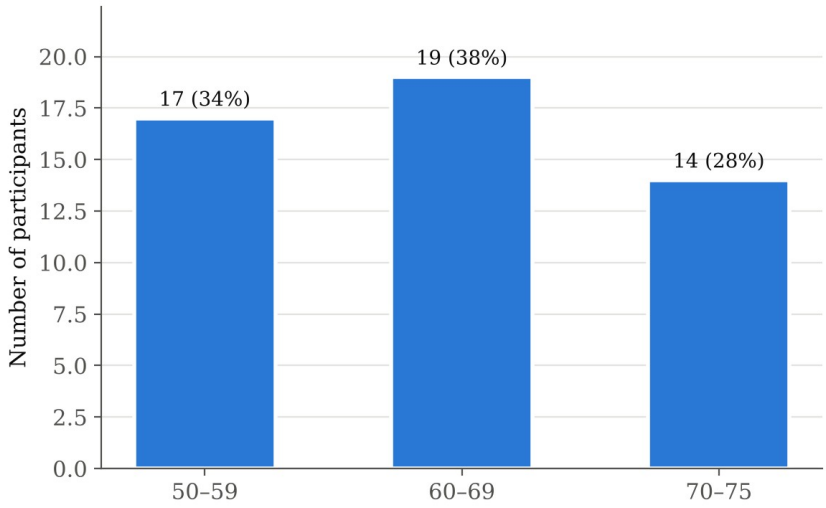


Table 6: Distribution of participants by BMI category

BMI category (kg/m²)	Frequency (n)	Percentage (%)
Normal (18.5–24.9)	15	30.0
Overweight (25–29.9)	26	52.0
Obese (≥30)	9	18.0
Total	50	100.0

More than half of the participants were overweight (26 (52.0%)), 9 (18.0%) were obese and 15 (30.0%) had a normal BMI (Graph 4).

Graph 4: Distribution of participants by BMI category

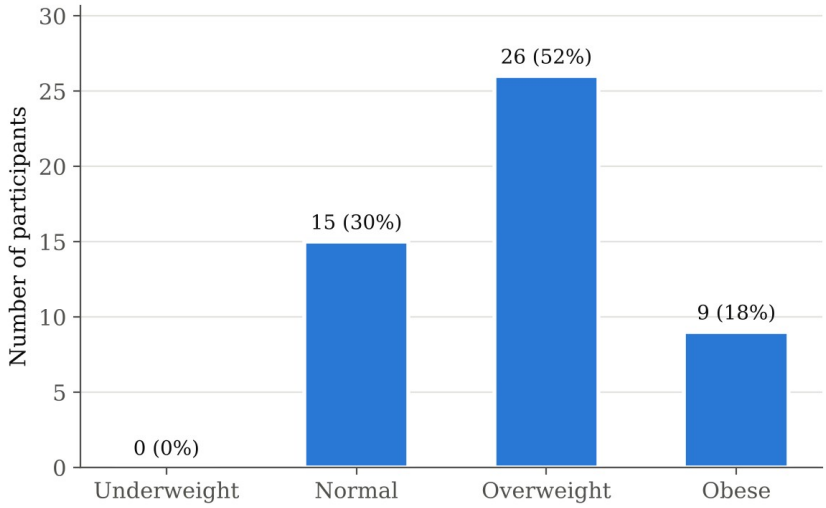


Table 7: Distribution of participants by post-operative duration

Post-operative duration	Frequency (n)	Percentage (%)
4–8 weeks	15	30.0
9–12 weeks	8	16.0
13–24 weeks	27	54.0
Total	50	100.0

2. Outcome measures

Table 8: Descriptive statistics of TSK-17 score and Stair Climb Test time

Outcome measure	Mean ± SD	Median (IQR)	Minimum	Maximum	Std. Error
TSK-17 score	41.76 ± 6.54	42.0 (37.0–44.0)	26	57	0.92
SCT time (seconds)	20.11 ± 3.16	20.63 (17.88–22.27)	12.82	26.96	0.45

The mean TSK-17 score was 41.76 ± 6.54 (range 26–57), which lies above the cut-off of 37 for high kinesiophobia. The mean time taken to complete the Stair Climb Test was 20.11 ± 3.16 seconds (range 12.82–26.96 seconds) (Table 8, Graph 5).

Graph 5: Mean TSK-17 score and mean SCT time

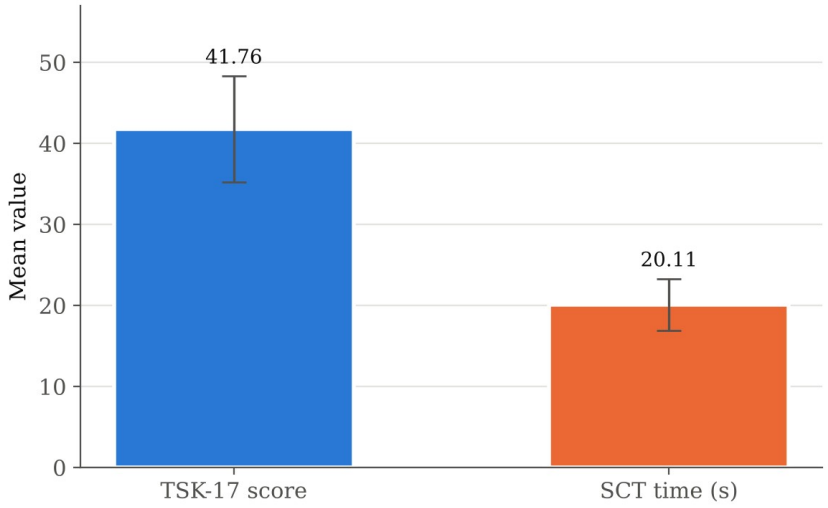


Table 9: Distribution of participants by level of kinesiophobia

Level of kinesiophobia	TSK-17 score	Frequency (n)	Percentage (%)
Low	17–37	14	28.0
High	38–68	36	72.0
Total		50	100.0

High kinesiophobia (TSK-17 > 37) was present in 36 (72.0%) participants, while 14 (28.0%) had low kinesiophobia (Graph 6).

Graph 6: Distribution of participants by level of kinesiophobia

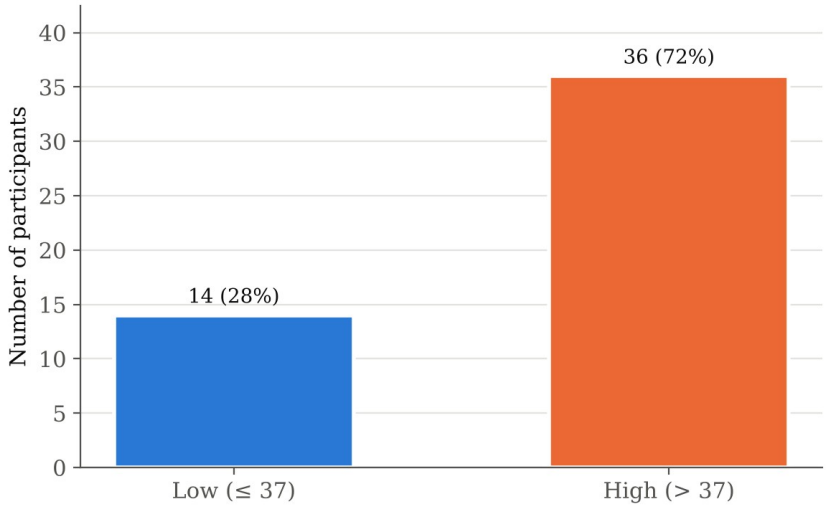
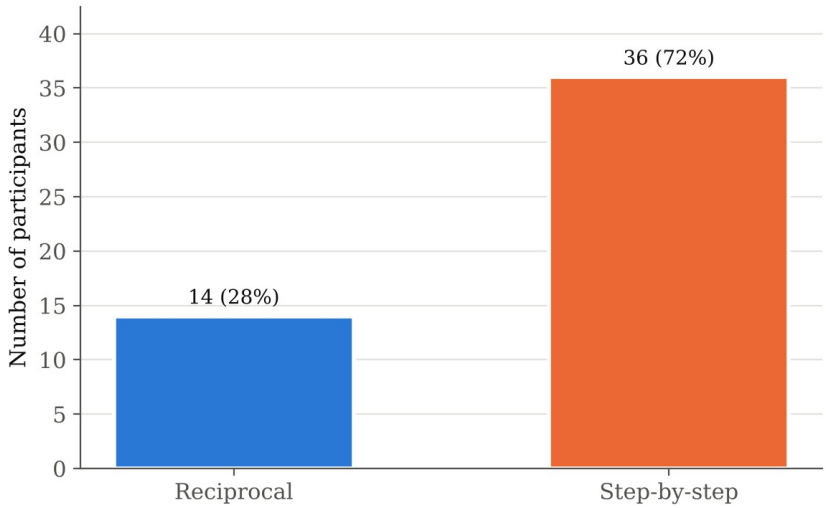


Table 10: Stair negotiation pattern, handrail support and assistive device

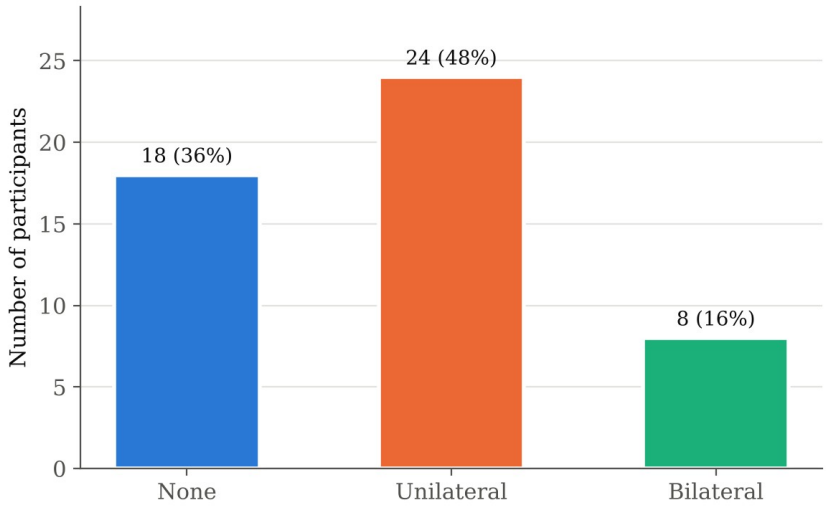
Variable	Category	Frequency (n)	Percentage (%)
Stair pattern	Reciprocal	14	28.0
	Step-by-step	36	72.0
Handrail support	None	18	36.0
	Unilateral	24	48.0
	Bilateral	8	16.0
Assistive device	None	25	50.0
	Single Cane	19	38.0
	Elbow Crutch	3	6.0
	Walker	3	6.0

A step-by-step (non-reciprocal) stair pattern was used by 36 (72.0%) participants, and only 14 (28.0%) negotiated the stairs reciprocally. 32 (64.0%) participants used the handrail (unilateral 24 (48.0%); bilateral 8 (16.0%)), and 25 (50.0%) were using an assistive device for ambulation, most commonly a single cane (19 (38.0%)) (Graphs 7–9).

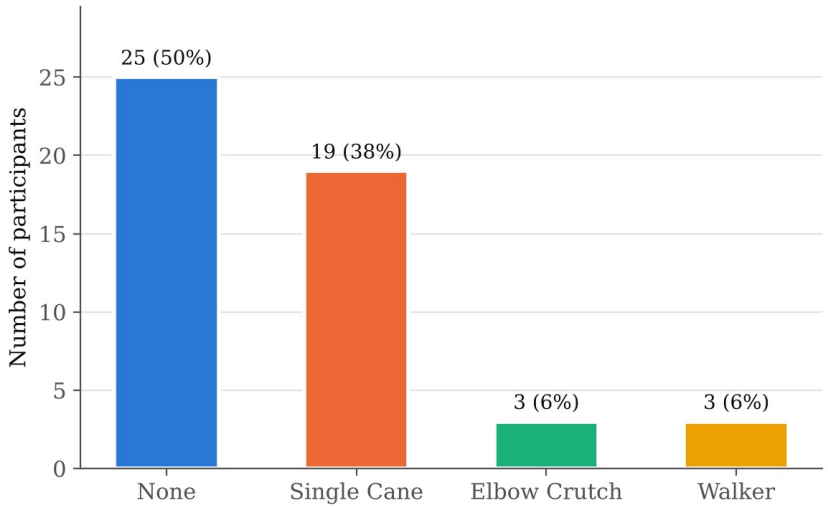
Graph 7: Stair negotiation pattern



Graph 8: Handrail support during the Stair Climb Test



Graph 9: Assistive device used for ambulation



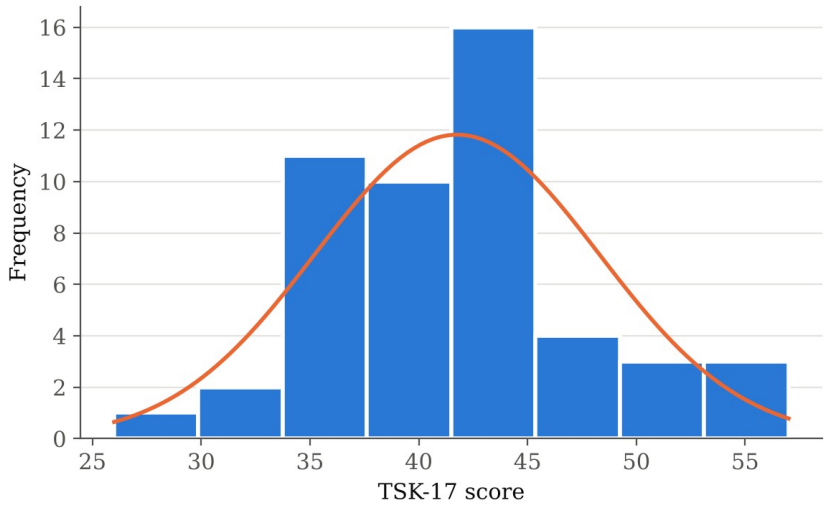
SUB-CHAPTER 2: TEST FOR NORMALITY

Normality of the continuous variables was assessed using the Shapiro–Wilk test. TSK-17 score (W = 0.966, p = 0.156), SCT time (W = 0.985, p = 0.766) and BMI were normally distributed (p > 0.05), whereas age and post-operative duration deviated from normality, reflecting their roughly uniform distribution across the inclusion range. As both primary variables were normally distributed, Pearson's correlation coefficient and parametric tests were used for the main analysis (Table 11; Graphs 10–13).

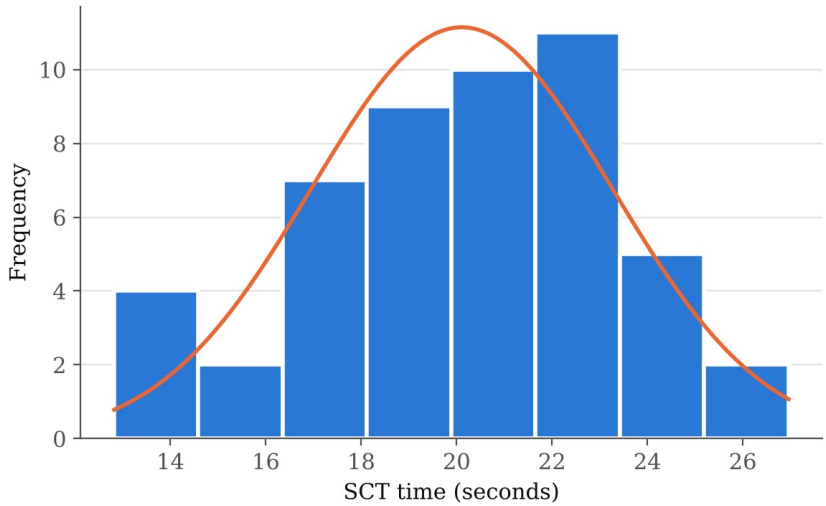
Table 11: Tests of normality (Shapiro–Wilk)

Variable	Statistic (W)	df	Sig. (p)	Distribution
Age	0.939	50	0.012	Not normal
BMI	0.969	50	0.214	Normal
Post-operative duration	0.931	50	0.006	Not normal
TSK-17 score	0.966	50	0.156	Normal
SCT time	0.985	50	0.766	Normal

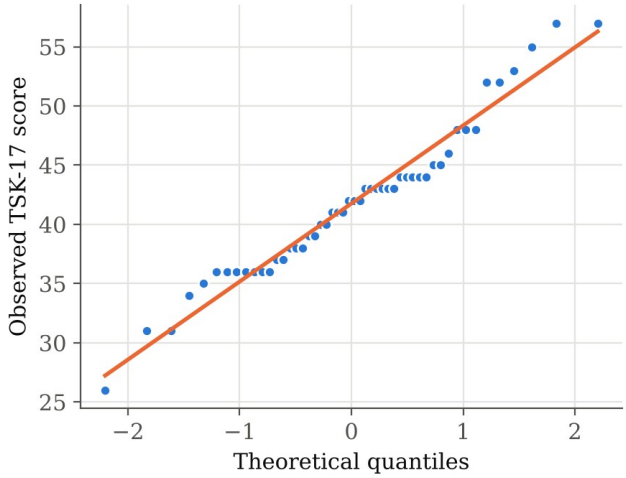
Graph 10: Histogram of TSK-17 scores with normal curve



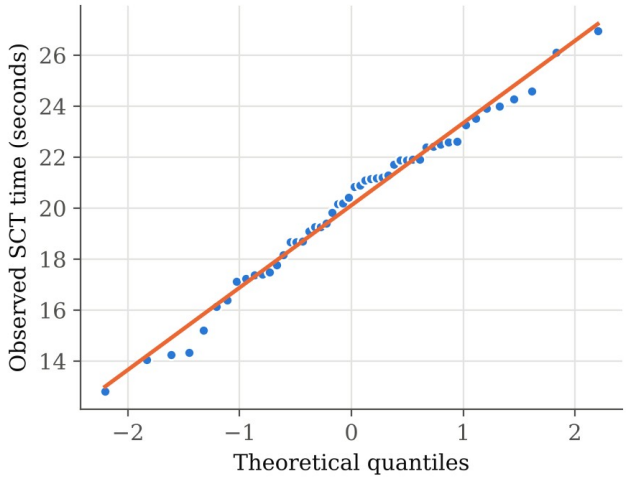
Graph 11: Histogram of SCT time with normal curve



Graph 12: Normal Q–Q plot of TSK-17 scores



Graph 13: Normal Q–Q plot of SCT time



SUB-CHAPTER 3: CORRELATION BETWEEN KINESIOPHOBIA AND STAIR-CLIMBING ABILITY

Table 12: Pearson correlation between TSK-17 score and SCT time

Variables	N	Pearson r	95% CI of r	R^2	p-value	Interpretation
TSK-17 score vs SCT time	50	0.685	0.503 to 0.809	0.470	< 0.001	Moderate positive, significant

Pearson's correlation analysis showed a **moderate, positive and statistically significant correlation** between TSK-17 score and SCT time ($r = 0.685$, 95% CI 0.503–0.809, $p < 0.001$). The coefficient of determination ($R^2 = 0.470$) indicates that 47.0% of the variance in stair-climbing time was explained by kinesiophobia. Spearman's rank correlation gave a

consistent result (ρ = 0.673, p < 0.001). Since time is the measure of the SCT, a positive correlation means that participants with greater fear of movement took longer to climb the stairs, i.e. had poorer stair-climbing ability (Graph 14). **The null hypothesis (H_0) is therefore rejected and the alternative hypothesis (H_1) is accepted.**

Graph 14: Scatter plot showing the correlation between TSK-17 score and SCT time

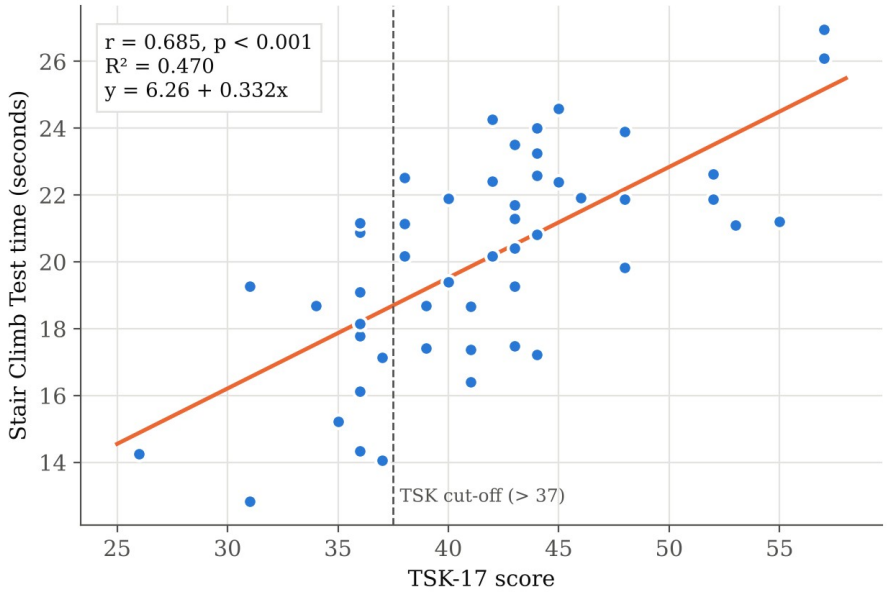


Table 13: Simple linear regression of SCT time on TSK-17 score

Model	B (unstandardised)	Std. Error	β (standardised)	p-value
Constant	6.261	–	–	–
TSK-17 score	0.332	0.051	0.685	< 0.001

The regression equation was **SCT time (s) = 6.26 + 0.332 × TSK-17 score**. Each one-point increase in TSK-17 score was associated with an increase of 0.33 seconds in SCT time, and a 10-point increase with approximately 3.3 seconds.

SUB-CHAPTER 4: COMPARISON OF STAIR-CLIMBING ABILITY BETWEEN LOW AND HIGH KINESIOPHOBIA GROUPS

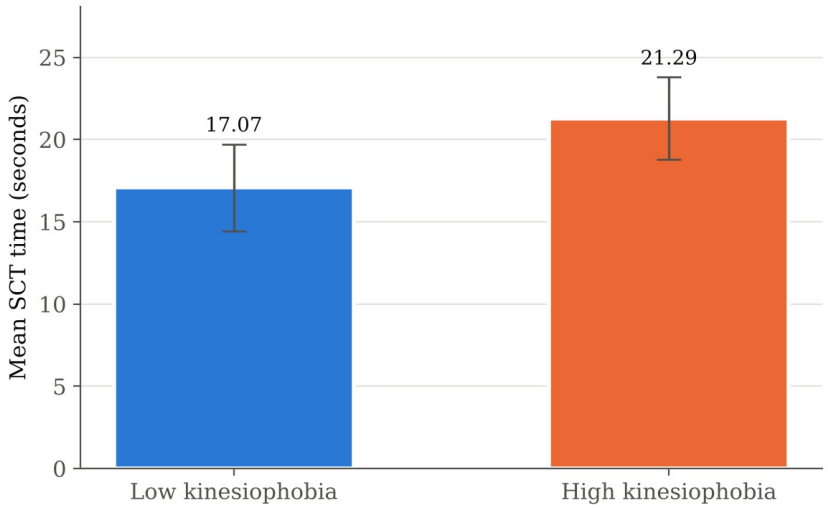
Table 14: Comparison of TSK-17 score and SCT time between low and high kinesiophobia groups (independent t-test)

Variable	Group	Mean ± SD	t	p-value	Result
TSK-17 score	Low (n = 14)	34.50 ± 3.11	-6.78	< 0.001	Significant
	High (n = 36)	44.58 ± 5.20			

Variable	Group	Mean ± SD	t	p-value	Result
SCT time (s)	Low (n = 14)	17.07 ± 2.65	-5.27	< 0.001	Significant
	High (n = 36)	21.29 ± 2.50			

Participants with high kinesiophobia took significantly longer to complete the SCT (21.29 ± 2.50 s) than those with low kinesiophobia (17.07 ± 2.65 s); mean difference 4.22 s, t(48) = 5.27, p < 0.001. The effect size was large (Cohen's d = 1.66). Levene's test confirmed equal variances (p = 0.571) (Graph 15).

Graph 15: Mean SCT time in low and high kinesiophobia groups



SUB-CHAPTER 5: COMPARISON ACROSS DEMOGRAPHIC AND CLINICAL SUBGROUPS

Table 15: Comparison of TSK-17 score and SCT time by gender, affected side and stair pattern (independent t-test)

Variable	Group	Mean ± SD	t	p-value	Result
TSK-17 by gender	Male (n = 20)	41.75 ± 6.43	-0.01	0.993	Not significant
	Female (n = 30)	41.77 ± 6.72			
SCT by gender	Male (n = 20)	20.52 ± 3.06	0.74	0.463	Not significant
	Female (n = 30)	19.84 ± 3.25			
TSK-17 by side	Right (n = 23)	42.17 ± 7.47	0.41	0.684	Not significant

Variable	Group	Mean ± SD	t	p-value	Result
	Left (n = 27)	41.41 ± 5.75			
SCT by side	Right (n = 23)	19.24 ± 3.65	-1.84	0.072	Not significant
	Left (n = 27)	20.85 ± 2.52			
TSK-17 by stair pattern	Reciprocal (n = 14)	42.14 ± 7.87	0.26	0.799	Not significant
	Step-by-step (n = 36)	41.61 ± 6.06			
SCT by stair pattern	Reciprocal (n = 14)	19.98 ± 3.21	-0.17	0.862	Not significant
	Step-by-step (n = 36)	20.16 ± 3.19			

TSK-17 scores and SCT times did not differ significantly between male and female participants, between right and left TKR, or between participants using a reciprocal and a step-by-step stair pattern (all $p > 0.05$). Participants with left TKR tended to be slower than those with right TKR (20.85 vs 19.24 s), but the difference was not significant (Mann–Whitney $U = 231.0$, $p = 0.124$; Levene's $p = 0.027$).

Table 16: Comparison of TSK-17 score and SCT time by handrail support and assistive device (one-way ANOVA)

Variable	Group	Mean ± SD	F	p-value	Result
TSK-17 by handrail	None (n = 18)	41.78 ± 8.24	0.12	0.887	Not significant
	Unilateral (n = 24)	41.42 ± 4.58			
	Bilateral (n = 8)	42.75 ± 8.01			
SCT by handrail	None (n = 18)	19.22 ± 3.32	2.87	0.067	Not significant
	Unilateral (n = 24)	20.04 ± 3.07			
	Bilateral (n = 8)	22.32 ± 2.20			
TSK-17 by device	None (n = 25)	41.48 ± 6.46	0.85	0.475	Not significant
	Single Cane (n = 19)	41.89 ± 6.29			

Variable	Group	Mean ± SD	F	p-value	Result
	Elbow Crutch (n = 3)	46.67 ± 4.62			
	Walker (n = 3)	38.33 ± 10.69			
SCT by device	None (n = 25)	20.11 ± 3.10	0.11	0.956	Not significant
	Single Cane (n = 19)	19.90 ± 3.10			
	Elbow Crutch (n = 3)	20.81 ± 3.10			
	Walker (n = 3)	20.70 ± 5.63			

Neither TSK-17 score nor SCT time differed significantly according to handrail support or assistive device ($p > 0.05$). Participants using bilateral handrail support showed the longest SCT time (22.32 ± 2.20 s), with a trend towards significance ($F = 2.87$, $p = 0.067$).

Table 17: Correlation of age, BMI and post-operative duration with TSK-17 score and SCT time

Variable	r with TSK-17	p-value	r with SCT time	p-value
Age	-0.106	0.463	-0.067	0.645
BMI	-0.235	0.101	-0.113	0.436
Post-operative duration	0.054	0.707	0.134	0.355

Age, BMI and post-operative duration showed only negligible, non-significant correlations with both TSK-17 score and SCT time (all $|r| < 0.30$, $p > 0.05$), indicating that these variables did not confound the main association.

Table 18: Association of kinesiophobia level with stair pattern and handrail support (chi-square test)

Variable	Category	Low kinesiophobia n	High kinesiophobia n	χ^2 (df)	p-value
Stair pattern	Reciprocal	3	11	0.087 (1)	0.768
	Step-by-step	11	25		
Handrail support	None	6	12	1.224 (2)	0.542
	Unilateral	5	19		
	Bilateral	3	5		

There was no significant association between the level of kinesiophobia and stair pattern ($\chi^2 = 0.087$, $p = 0.768$) or handrail support ($\chi^2 = 1.224$, $p = 0.542$).

Table 19: Hypothesis test summary

Null hypothesis	Test	Sig.	Decision
There is no significant correlation between kinesiophobia (TSK-17) and stair-climbing ability (SCT) following unilateral TKR.	Pearson correlation	< 0.001	Reject the null hypothesis
SCT time is the same in participants with low and high kinesiophobia.	Independent-samples t-test	< 0.001	Reject the null hypothesis

CHAPTER 12. DISCUSSION

DISCUSSION

The present cross-sectional study examined the correlation between fear of movement (kinesiophobia) and stair-climbing ability in 50 individuals (30 female, 20 male; mean age 63.50 ± 7.05 years) assessed 4–24 weeks after unilateral Total Knee Replacement. Kinesiophobia was measured with the TSK-17 and stair-climbing ability with the timed Stair Climb Test. The principal finding was a moderate, positive and statistically significant correlation between TSK-17 score and SCT time ($r = 0.685$, $p < 0.001$), with kinesiophobia explaining 47% of the variance in stair-climbing time. Participants with high kinesiophobia took on average 4.22 seconds longer to complete the SCT than those with low kinesiophobia ($p < 0.001$, Cohen's $d = 1.66$). The null hypothesis was therefore rejected.

12.1 Level of Kinesiophobia after Unilateral TKR

The mean TSK-17 score in this study was 41.76 ± 6.54, and 72% of participants had high kinesiophobia (TSK-17 > 37). The mean score is comparable to the mean TSK score of 40.47 ± 7.01 reported by Özdemir and Karazeybek (2024)[41] in 205 patients after TKR, but the proportion with high kinesiophobia is considerably higher than the pooled prevalence of 35% reported in the meta-analysis of Du et al. (2025)[43] and the 27% reported by Aleksić et al. (2024)[38] on the fifth post-operative day.

Several factors may explain the high prevalence in the present sample. First, all participants had undergone unilateral TKR: a patient with one healthy, pain-free limb can continually offload the operated knee, so that fearful beliefs are never disconfirmed by successful loading. Second, participants were assessed between 4 and 24 weeks after surgery, the period in which Deniz et al. (2022)[36] found kinesiophobia to peak (at three months) before declining by 12 months. Third, Du et al. (2025)[43] identified post-operative pain, low social support, negative coping, low education and lower income as risk factors, and Özdemir and Karazeybek (2024)[41] linked older age and lower education with greater fear of movement; such factors, together with limited pre-operative counselling and access to structured physiotherapy, are common in the Indian setting. Differences in the TSK version and cut-off used across studies may also contribute to variation in reported prevalence.

12.2 Stair-Climbing Ability after Unilateral TKR

The mean SCT time was 20.11 ± 3.16 seconds. Stair negotiation remained clearly impaired: 72% of participants used a non-reciprocal step-by-step pattern, 64% relied on handrail support and half were still using an assistive device for ambulation. These findings are consistent with Murakami et al. (2024)[37], who reported frequent use of step-by-step patterns and handrails and longer stair times three months after TKR, particularly during descent, and

with Zeni and Snyder-Mackler (2010b)[52], who found that many patients continued to use handrails up to two years after surgery. Lee et al. (2020)[35] showed that stair-climbing ability one month after TKA depends on pre-operative balance and quadriceps strength, and Rasnick et al. (2016)[31] demonstrated reduced quadriceps forces with compensatory flexor activity during stair ascent after TKR, reflecting the high neuromuscular demand of stair negotiation.

12.3 Correlation between Kinesiophobia and Stair-Climbing Ability

The moderate positive correlation (r = 0.685) between TSK-17 score and SCT time supports the fear-avoidance model[13,19], which proposes that catastrophic interpretation of pain leads to fear of movement, avoidance and hypervigilance and, through disuse, to reduced physical performance. The regression analysis showed that each one-point increase in TSK-17 score was associated with an increase of 0.33 seconds in SCT time; thus a patient scoring 50 would be expected to take roughly 5.0 seconds longer than a patient scoring 35, a difference of about 25% of the mean SCT time.

These findings are in agreement with Güney-Deniz et al. (2017)[32], who reported that kinesiophobia explained 41% of the variance in walking distance and 47% of the variance in knee flexion early after TKA, with Aleksić et al. (2024)[38], who found that patients with kinesiophobia walked more slowly and were more dependent, and with Deniz et al. (2022)[36], who reported negative correlations between kinesiophobia and spatiotemporal gait parameters three months after surgery. In two prospective cohorts, kinesiophobia was shown to slow recovery and independently predict poorer outcome after TKA[30,33], and the systematic review of Brown et al. (2020)[34] concluded that kinesiophobia impairs function for up to one year. The present study extends this evidence to a specific, high-demand task – stair negotiation – measured with an objective performance test, in an Indian population.

Several mechanisms may explain how fear of movement slows stair negotiation. First, fear directly alters motor control: Asaki and Anan (2025)[42] documented abnormal activity of the semitendinosus and medial gastrocnemius and a prolonged swing phase during stair descent in a patient with persistent fear, despite improved range of motion and self-efficacy. Protective co-contraction around the knee limits the knee flexion required for smooth descent and slows the movement. Second, fearful patients tend to avoid stairs and exercise, which may perpetuate quadriceps weakness – an established determinant of stair ability[35]. Third, in unilateral TKR the healthy limb allows consistent offloading of the operated knee, so that eccentric control of the replaced knee is not re-learned.

Interestingly, although SCT time was strongly related to kinesiophobia, the categorical stair strategies were not: stair pattern and handrail use did not differ significantly between low and

high kinesiophobia groups. This suggests that strategies such as a step-by-step pattern or handrail use were adopted by most participants at this stage of recovery regardless of fear level, perhaps as a habitual safety behaviour encouraged during in-patient rehabilitation, whereas the timed SCT was more sensitive to the hesitancy and slowness associated with fear. Participants using bilateral handrail support tended to be slowest (22.32 s), although this trend did not reach significance.

Age, BMI and post-operative duration were not significantly correlated with either kinesiophobia or SCT time, and neither outcome differed by gender or operated side. The absence of an association with post-operative duration contrasts with longitudinal studies showing a decline in kinesiophobia over the post-operative months[36,44]. Because the present design was cross-sectional, between-subject variability in fear may have outweighed the effect of time since surgery. These results indicate that fear of movement was associated with stair-climbing ability independently of these demographic factors.

Physical factors such as quadriceps strength, balance and pre-operative function are known predictors of post-operative stair ability[35,52]. The finding that kinesiophobia alone explained almost half of the variance in SCT time highlights that psychological factors deserve the same attention as physical impairments in post-operative rehabilitation.

12.4 Clinical Implications

- Kinesiophobia should be screened routinely after TKR using the TSK-17, particularly in patients who are slow to progress with stair training[43].
- Because post-operative pain is the strongest predictor of kinesiophobia[38,43], effective pain management and early education should form part of rehabilitation.
- Cognitive behavioural approaches reduce kinesiophobia after TKA[39] and can be combined with conventional exercise.
- Graded exposure to stair negotiation – progressing step height, number of steps, reduced handrail use and reciprocal patterns – can help disconfirm fearful beliefs while building strength and motor control, with particular emphasis on stair descent[37].
- Prediction tools such as the nomogram developed by Xie et al. (2024)[40] may help identify high-risk patients for early intervention.

CHAPTER 13. CONCLUSION

CONCLUSION

The present study concludes that there is a **moderate, positive and statistically significant correlation** between fear of movement (kinesiophobia) and stair-climbing time in individuals following unilateral Total Knee Replacement ($r = 0.685$, $p < 0.001$). Higher kinesiophobia was associated with poorer stair-climbing ability, and kinesiophobia explained 47% of the variance in Stair Climb Test time. Participants with high kinesiophobia took significantly longer to negotiate stairs than those with low kinesiophobia. The alternative hypothesis is therefore accepted.

High kinesiophobia was present in 72% of participants, indicating that fear of movement is very common in the early months after unilateral TKR. Kinesiophobia should therefore be routinely assessed and addressed alongside physical impairments in post-operative rehabilitation to improve stair negotiation and functional independence.

CHAPTER 14. ETHICAL CONSIDERATION

ETHICAL CONSIDERATION

Ethical approval was obtained from the Institutional Ethics Committee of [name of committee] (Ref. No. [_____], dated [__/__/____]) before the commencement of the study. The study was conducted in accordance with the Declaration of Helsinki[53] and the National Ethical Guidelines for Biomedical and Health Research Involving Human Participants of the Indian Council of Medical Research (ICMR)[54].

- Written informed consent was obtained from every participant after the purpose, procedure, risks and benefits of the study had been explained in a language they understood.
- Participation was voluntary, and participants were free to withdraw at any time without any effect on their treatment.
- Confidentiality was maintained by assigning each participant a code (TKR-2025-001 to TKR-2025-050); no identifying information is reported.
- The assessment was non-invasive. Participants were guarded during the Stair Climb Test, rest was allowed, and the test was stopped immediately if pain, dizziness or breathlessness occurred.
- No financial burden was placed on the participants and no incentives were offered.

CHAPTER 15. LIMITATION AND SCOPE OF FUTURE STUDY

LIMITATION AND SCOPE OF FUTURE STUDY

Limitations

1. The cross-sectional design does not allow causal inference; it cannot determine whether fear causes poor stair performance or poor performance increases fear.
2. Convenience sampling from [a single centre] in Jaipur limits the generalisability of the findings.
3. The post-operative window (4–24 weeks) was wide, and both kinesiophobia and stair ability change during this period.
4. Potential confounders such as pain intensity, quadriceps strength, knee range of motion, pre-operative function, anxiety, depression and pain catastrophising were not measured.
5. Only total SCT time was recorded; ascent and descent were not timed separately, and kinematic or EMG data were not collected.
6. Kinesiophobia was assessed with a self-report questionnaire, which is subject to response bias.

Scope of Future Study

- Longitudinal studies following patients from before surgery to one year after TKR to examine how changes in kinesiophobia relate to changes in stair performance.
- Larger, multi-centre studies comparing unilateral and bilateral TKR.
- Multivariate models including pain, quadriceps strength, range of motion, self-efficacy and catastrophising to determine the independent contribution of kinesiophobia.
- Separate timing of stair ascent and descent, with motion analysis or surface EMG to examine fear-related movement patterns.
- Randomised controlled trials of interventions targeting kinesiophobia (education, graded exposure, cognitive-behavioural therapy) on stair-climbing outcomes.

CHAPTER 16. BIBLIOGRAPHY

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CHAPTER 17. ANNEXURE

ANNEXURE

ANNEXURE I: ASSESSMENT PROFORMA FOR THE SUBJECTS

SUBJECT ID: _____ DATE: ____/____/_____

Item	Details
Name	
Age (years)	
Gender	Male / Female
Height (cm)	
Weight (kg)	
BMI (kg/m²)	
Occupation	
Residence	Single storey / Multi-storey; stairs at home: Yes / No
Date of surgery	
Post-operative duration (weeks)	
Operated side	Right / Left
Assistive device	None / Single cane / Elbow crutch / Walker
Currently receiving physiotherapy	Yes / No
Relevant medical history	
Inclusion criteria met	Yes / No
Exclusion criteria absent	Yes / No

ANNEXURE II: PARTICIPANT INFORMATION SHEET AND CONSENT FORM

Study title: "Correlation between Fear of Movement (Kinesiophobia) and Stair-Climbing Ability following Unilateral Total Knee Replacement: A Cross-Sectional Study"

Investigator: Kshma Sahu, MPT student, Department of Physiotherapy, Suresh Gyan Vihar University, Jaipur. Contact: [phone number]

Guide: Dr. Sheenam Popli, Associate Professor, Department of Physiotherapy, Suresh Gyan Vihar University, Jaipur.

You are invited to take part in a study that aims to find out whether fear of movement is related to the ability to climb stairs after knee replacement surgery on one knee. If you agree, you will answer some questions about yourself and your surgery, complete a 17-item questionnaire about movement and pain (about 10 minutes), and walk up and down a flight of stairs at a fast but safe pace while being timed. You may use the handrail and your usual walking aid. A physiotherapist will stay beside you throughout. Stair climbing may cause mild, temporary knee discomfort or tiredness; you may rest or stop at any time. Your information will be kept confidential and used only for research. Participation is voluntary, and you may withdraw at any time without any effect on your treatment.

CONSENT FORM

1. I confirm that I have read (or have had read to me) and understood the information sheet for the above study and have had the opportunity to ask questions.
2. I understand that my participation is voluntary and that I am free to withdraw at any time without giving any reason and without my medical care being affected.
3. I understand that my personal information will be kept confidential and that the data may be used for research and publication without revealing my identity.
4. I agree to take part in the above study.

Name of participant: _____

Signature / thumb impression: _____

Date: _____

Name and signature of witness: _____

Name and signature of investigator:

_____

सूचित सहमति पत्र

1. मैंने उपरोक्त अध्ययन की प्रतिभागी सूचना पत्र को पढ़ लिया है (या मुझे पढ़कर सुनाया गया है) और समझ लिया है, तथा मुझे प्रश्न पूछने का अवसर दिया गया है।
2. मैं समझता/समझती हूँ कि मेरी भागीदारी स्वैच्छिक है और मैं किसी भी समय, बिना कोई कारण बताए, अध्ययन से हट सकता/सकती हूँ, और इससे मेरे उपचार पर कोई प्रभाव नहीं पड़ेगा।
3. मैं समझता/समझती हूँ कि मेरी व्यक्तिगत जानकारी गोपनीय रखी जाएगी और मेरी पहचान उजागर किए बिना आँकड़ों का उपयोग शोध एवं प्रकाशन के लिए किया जा सकता है।
4. मैं इस अध्ययन में भाग लेने के लिए सहमत हूँ।

प्रतिभागी का नाम: _____

हस्ताक्षर / अँगूठे का निशान: _____

दिनांक: _____

गवाह का नाम एवं हस्ताक्षर: _____

शोधकर्ता का नाम एवं हस्ताक्षर: _____

ANNEXURE III: DATA COLLECTION FORM

Date: _____ Name: _____ Age/Gender: _____

Subject ID: _____

TAMPA SCALE FOR KINESIOPHOBIA (TSK-17)

Please indicate how much you agree or disagree with each statement by ticking one box. Think about how you feel now about your operated knee.

No.	Statement	Strongly disagree (1)	Disagree (2)	Agree (3)	Strongly agree (4)
1	I'm afraid that I might injure myself if I exercise.	☐	☐	☐	☐
2	If I were to try to overcome it, my pain would increase.	☐	☐	☐	☐
3	My body is telling me I have something dangerously wrong.	☐	☐	☐	☐
4 *	My pain would probably be relieved if I were to exercise.	☐	☐	☐	☐
5	People aren't taking my medical condition seriously enough.	☐	☐	☐	☐
6	My accident has put my body at risk for the rest of my life.	☐	☐	☐	☐
7	Pain always means I have injured my body.	☐	☐	☐	☐
8 *	Just because something aggravates my pain does not mean it is dangerous.	☐	☐	☐	☐
9	I am afraid that I might injure myself accidentally.	☐	☐	☐	☐
10	Simply being careful that I do not make any unnecessary movements is the safest thing I can do to prevent my pain from worsening.	☐	☐	☐	☐
11	I wouldn't have this much pain if there weren't something potentially dangerous going on in my body.	☐	☐	☐	☐
12 *	Although my condition is painful, I would be better off if I were physically active.	☐	☐	☐	☐

No.	Statement	Strongly disagree (1)	Disagree (2)	Agree (3)	Strongly agree (4)
13	Pain lets me know when to stop exercising so that I don't injure myself.	☐	☐	☐	☐
14	It's really not safe for a person with a condition like mine to be physically active.	☐	☐	☐	☐
15	I can't do all the things normal people do because it's too easy for me to get injured.	☐	☐	☐	☐
16*	Even though something is causing me a lot of pain, I don't think it's actually dangerous.	☐	☐	☐	☐
17	No one should have to exercise when he/she is in pain.	☐	☐	☐	☐

Reverse-scored items (4, 8, 12, 16). Total score range 17–68; > 37 = high kinesiophobia. TSK-17 total: _____

STAIR CLIMB TEST – RECORDING SHEET

Number of steps: _____ Step height: _____ cm Handrails: one side / both sides

Parameter	Observation
SCT time (seconds)	
Stair pattern	Reciprocal / Step-by-step
Handrail support	None / Unilateral / Bilateral
Assistive device	None / Single cane / Elbow crutch / Walker
Remarks / adverse events	

ANNEXURE IV: ETHICAL CLEARANCE CERTIFICATE

Insert a scanned copy of the Institutional Ethics Committee approval letter here.

ANNEXURE V: PLAGIARISM REPORT

Insert the plagiarism check certificate issued by the University library here.

CHAPTER 18. MASTER CHART

MASTER CHART

MASTER CHART: CORRELATION BETWEEN KINESIOPHOBIA AND STAIR-CLIMBING ABILITY FOLLOWING UNILATERAL TKR

Table 20: Master chart of all participants (n = 50)

S. No.	Patient ID	Age	Gender	BMI	Side	Post-op (wks)	TSK-17	SCT (s)	Handrail	Stair pattern	Assistive device
1	TKR-001	57	M	26.2	L	7	44	22.58	Unilateral	Recip.	Elbow Crutch
2	TKR-002	70	M	27.0	R	6	37	17.13	Unilateral	Step	None
3	TKR-003	65	M	20.6	R	20	36	16.13	None	Step	None
4	TKR-004	61	F	24.2	L	20	43	21.29	None	Step	None
5	TKR-005	58	F	20.1	L	15	43	17.49	None	Step	Single Cane
6	TKR-006	71	F	25.7	L	17	38	22.51	None	Step	None
7	TKR-007	57	M	26.9	L	24	38	21.14	Bilateral	Recip.	None
8	TKR-008	69	M	23.9	R	9	43	19.27	Unilateral	Recip.	Single Cane
9	TKR-009	73	M	34.9	R	6	31	12.82	Unilateral	Step	None
10	TKR-010	61	F	27.2	L	12	31	19.26	None	Step	Single Cane
11	TKR-011	61	F	30.0	L	8	36	17.78	Unilateral	Step	None
12	TKR-012	74	M	28.6	L	20	40	21.90	Unilateral	Step	None
13	TKR-013	71	F	21.0	R	17	44	17.23	None	Step	Elbow Crutch
14	TKR-014	54	M	26.8	L	24	52	21.88	Unilateral	Step	None
15	TKR-015	58	F	24.7	L	6	48	21.87	Unilateral	Recip.	None

S. No.	Patient ID	Age	Gender	BMI	Side	Post-op (wks)	TSK-17	SCT (s)	Handrail	Stair pattern	Assistive device
16	TKR-016	74	F	28.7	L	4	40	19.40	Unilateral	Step	Single Cane
17	TKR-017	53	M	26.1	L	23	41	18.67	Unilateral	Step	Single Cane
18	TKR-018	72	F	24.1	L	24	34	18.69	None	Step	None
19	TKR-019	71	F	30.2	L	4	41	17.38	Unilateral	Step	Single Cane
20	TKR-020	52	M	26.5	R	6	39	18.68	Unilateral	Step	None
21	TKR-021	74	F	20.9	L	21	44	20.83	Unilateral	Recip.	None
22	TKR-022	62	F	27.2	L	13	36	19.10	Bilateral	Step	Single Cane
23	TKR-023	56	F	20.2	R	6	45	22.39	Unilateral	Step	Single Cane
24	TKR-024	52	F	29.0	R	11	53	21.10	None	Step	Single Cane
25	TKR-025	71	F	24.4	R	17	41	16.40	None	Recip.	Single Cane
26	TKR-026	51	F	28.6	R	21	44	23.26	Unilateral	Step	Walker
27	TKR-027	62	F	24.1	L	18	46	21.92	None	Step	Single Cane
28	TKR-028	72	F	32.1	R	5	39	17.41	Unilateral	Recip.	None
29	TKR-029	62	F	34.0	R	13	43	23.52	Bilateral	Step	Single Cane
30	TKR-030	67	M	22.3	R	5	42	20.18	Unilateral	Recip.	None
31	TKR-031	60	M	28.9	R	20	42	22.42	Unilateral	Recip.	None
32	TKR-032	66	M	29.4	R	11	36	18.16	Unilateral	Step	None
33	TKR-033	65	M	27.2	L	4	42	24.27	Unilateral	Step	None

S. No.	Patient ID	Age	Gender	BMI	Side	Post-op (wks)	TSK-17	SCT (s)	Handrail	Stair pattern	Assistive device
34	TKR-034	65	F	25.5	L	12	45	24.59	Unilateral	Step	Walker
35	TKR-035	69	M	25.9	R	14	52	22.62	Bilateral	Step	Elbow Crutch
36	TKR-036	62	M	30.2	L	19	48	23.90	Unilateral	Step	Single Cane
37	TKR-037	73	M	26.5	R	10	57	26.10	Bilateral	Recip.	None
38	TKR-038	70	M	26.3	L	13	36	20.89	Bilateral	Recip.	None
39	TKR-039	53	F	20.4	R	6	48	19.83	None	Step	None
40	TKR-040	55	F	37.2	L	21	43	21.70	None	Step	None
41	TKR-041	69	F	26.2	L	16	57	26.96	None	Step	Single Cane
42	TKR-042	57	F	28.2	R	10	36	14.33	None	Step	Single Cane
43	TKR-043	71	F	33.8	L	7	35	15.22	None	Recip.	Single Cane
44	TKR-044	59	F	28.1	R	16	37	14.06	Unilateral	Step	None
45	TKR-045	57	F	29.3	R	23	26	14.25	None	Recip.	Walker
46	TKR-046	68	F	21.4	L	4	38	20.17	Unilateral	Step	Single Cane
47	TKR-047	54	M	28.8	L	11	36	21.17	Bilateral	Step	Single Cane
48	TKR-048	64	M	31.1	L	17	43	20.42	None	Step	Single Cane
49	TKR-049	68	F	25.6	R	19	44	24.00	Bilateral	Step	None
50	TKR-050	59	F	24.4	R	17	55	21.20	None	Recip.	None

M = Male, F = Female; R = Right, L = Left; Step = step-by-step; Recip. = reciprocal. Patient IDs abbreviated from TKR-2025-001 to TKR-050.

RAW DATA ANALYSIS

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation	Skewness	Kurtosis
Age	50	51.00	74.00	63.50	7.054	-0.118	-1.285
BMI	50	20.10	37.20	26.81	3.868	0.333	0.281
PostOp_Weeks	50	4.00	24.00	13.44	6.440	0.036	-1.317
TSK_Score	50	26.00	57.00	41.76	6.539	0.377	0.409
SCT_Time	50	12.82	26.96	20.11	3.163	-0.261	-0.218

Tests of Normality (Shapiro–Wilk)

	Statistic	df	Sig.
Age	.939	50	.012
BMI	.969	50	.214
PostOp_Weeks	.931	50	.006
TSK_Score	.966	50	.156
SCT_Time	.985	50	.766

Correlations

		TSK_Score	SCT_Time
TSK_Score	Pearson Correlation	1	.685**
	Sig. (2-tailed)		.000
	N	50	50
SCT_Time	Pearson Correlation	.685**	1
	Sig. (2-tailed)	.000	
	N	50	50

Correlation is significant at the 0.01 level (2-tailed

Group Statistics (SCT_Time by TSK group)

TSK group	N	Mean	Std. Deviation	Std. Error Mean
Low (≤ 37)	14	17.071	2.653	0.709
High (> 37)	36	21.291	2.503	0.417

Independent Samples Test (equal variances assumed)

	Levene's F	Levene's Sig.	t	df	Sig. (2-tailed)	Mean Difference	Cohen's d
SCT_Time	0.325	.571	-5.266	48	.000	-4.220	-1.659
TSK_Score	2.687	.108	-6.780	48	.000	-10.083	-2.135

Regression – Coefficients (Dependent variable: SCT_Time)

Model	B	Std. Error	Beta	Sig.
(Constant)	6.261			
TSK_Score	0.332	0.051	0.685	.000

Model summary: $R = 0.685$, $R^2 = 0.470$.

CHAPTER 19. PUBLICATION FORMAT

PUBLICATION FORMAT

CORRELATION BETWEEN FEAR OF MOVEMENT (KINESIOPHOBIA) AND STAIR-CLIMBING ABILITY FOLLOWING UNILATERAL TOTAL KNEE REPLACEMENT: A CROSS-SECTIONAL STUDY

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2. Associate Professor, Department of Physiotherapy, Suresh Gyan Vihar University, Jaipur

ABSTRACT

BACKGROUND AND PURPOSE

Despite successful Total Knee Replacement (TKR), many patients continue to have difficulty negotiating stairs. Fear of movement (kinesiophobia) may contribute to this limitation. The aim of this study was to determine the correlation between kinesiophobia and stair-climbing ability following unilateral TKR.

METHODOLOGY

A cross-sectional study was conducted on 50 individuals (aged 50–75 years) 4–24 weeks after unilateral TKR, selected by convenience sampling. Kinesiophobia was assessed with the Tampa Scale for Kinesiophobia (TSK-17) and stair-climbing ability with the timed Stair Climb Test (SCT). Pearson's correlation, linear regression and the independent t-test were used, with significance set at $p < 0.05$.

RESULT

The mean TSK-17 score was 41.76 ± 6.54, and 72% of participants had high kinesiophobia (TSK > 37). The mean SCT time was 20.11 ± 3.16 s. TSK-17 score correlated moderately and positively with SCT time ($r = 0.685$, $R^2 = 0.470$, $p < 0.001$). Participants with high kinesiophobia were 4.22 s slower than those with low kinesiophobia ($p < 0.001$, $d = 1.66$).

CONCLUSION

Higher kinesiophobia is associated with poorer stair-climbing ability after unilateral TKR. Routine screening for and management of fear of movement should be part of post-operative rehabilitation.

KEYWORDS

Kinesiophobia, Total Knee Replacement, Stair Climb Test, Tampa Scale for Kinesiophobia, Fear-avoidance

INTRODUCTION

Total Knee Replacement is the gold-standard treatment for end-stage knee osteoarthritis, yet about one in five patients remain dissatisfied after surgery[1]. Stair negotiation is among the most demanding and persistently impaired activities after TKR, with patients adopting step-to patterns and relying on handrails[2,3]. Physical factors such as quadriceps strength explain only part of this impairment[4]. According to the fear-avoidance model, catastrophic interpretation of pain leads to fear of movement, avoidance and disability[5]. Kinesiophobia affects about 35% of patients after TKA[6] and is associated with poorer function[7,8]. However, its relationship with objectively measured stair-climbing ability after unilateral TKR has rarely been studied, particularly in India.

METHODOLOGY

This cross-sectional study included 50 participants aged 50–75 years who were 4–24 weeks after primary unilateral TKR, recruited by convenience sampling from [study setting], Jaipur, after approval from the Institutional Ethics Committee. Patients with bilateral or revision TKR, post-operative complications, neurological, severe cardiopulmonary or other lower-limb conditions, or cognitive impairment were excluded. The sample size was calculated with G*Power ($\rho = 0.40$, $\alpha = 0.05$, power = 0.80)[9]. Kinesiophobia was measured with the TSK-17 (score 17–68; > 37 = high)[10] and stair-climbing ability with the timed SCT[11]. Data were analysed using Shapiro–Wilk, Pearson's correlation, linear regression and independent t-tests ($p < 0.05$).

RESULT

Participants had a mean age of 63.50 ± 7.05 years; 30 (60%) were female, and 27 (54%) had left TKR. The mean TSK-17 score was 41.76 ± 6.54 and the mean SCT time 20.11 ± 3.16 s; 72% used a step-by-step pattern and 64% used handrail support. Both variables were normally distributed. TSK-17 correlated positively with SCT time ($r = 0.685$, 95% CI 0.503–0.809, $p < 0.001$); SCT time = 6.26 + 0.332 × TSK. SCT time was 17.07 ± 2.65 s in the low and 21.29 ± 2.50 s in the high kinesiophobia group ($t = 5.27$, $p < 0.001$). Age, BMI, post-operative duration, gender and operated side were not significantly related to either outcome.

DISCUSSION

This study found that kinesiophobia explained almost half of the variance in stair-climbing time after unilateral TKR, consistent with the fear-avoidance model[5] and with previous reports linking kinesiophobia to poorer walking performance and range of motion[7]. Fear-related co-contraction and hesitancy during stair descent[12], avoidance-related weakness of the quadriceps, a known determinant of stair ability[4], and offloading onto the non-operated limb in unilateral TKR may explain the association. The high prevalence of kinesiophobia

(72%) exceeds the pooled estimate of 35%[6], possibly reflecting the early post-operative timing and unilateral surgery. Limitations include the cross-sectional design, convenience sampling and the absence of measures of pain and strength.

CONCLUSION

Kinesiophobia is common after unilateral TKR and shows a moderate positive correlation with stair-climbing time. Screening for and addressing fear of movement should be integrated into post-operative rehabilitation.

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